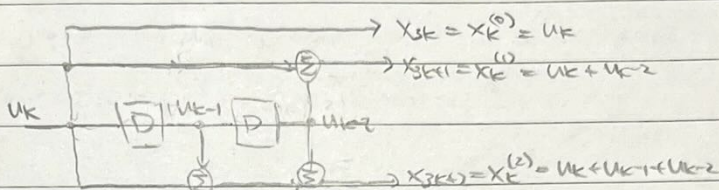


convolutional codes

convolutional codes

- Convolutional codes use a streaming encoder - generates n bits for every k bits, but is not a (n, k) block code since the ops depend on the current k bits and a number of past k bits.
- The rate of convolutional codes is $R = \frac{k}{n}$
- A convolutional code encoder consists of delay and sum operators. The n bits depend on the current and a few past k bits. This can be specified using octal notation.
- e.g. $(4, 5, 7)_8$ encoder - has connections $(100, 101, 111)$



Let $s_k = [u_k, u_{k-1}, u_{k-2}]$. For input sequence $u = [1, 0, 1, 0]$

$$k=1, \quad s_k = [1, 0, 0] \quad x_k = s_k \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^T = [1, 0, 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = [1, 1, 1]$$

$$k=2, \quad s_k = [0, 1, 0] \quad x_k = s_k \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^T = [0, 1, 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = [0, 0, 1]$$

$$k=3, \quad s_k = [1, 0, 1] \quad x_k = s_k \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^T = [1, 0, 1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = [1, 0, 0]$$

$$k=4, \quad s_k = [0, 1, 0] \quad x_k = s_k \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^T = [0, 1, 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = [0, 0, 1]$$

→ output sequence $x = [1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 1]$

+ If the octal number has more than 1 digit, we separate w/ ', e.g. $(34, 27)_8 \rightarrow$ connections $(011100, 010111)$ add leading zeros if needed.

convolutional codes with $R \neq \frac{1}{n}$.

- Many textbooks introduce generalised (n, k) convolutional codes ($R = \frac{k}{n}$), but in practice this is not used → very complex decoders.
- We can use puncturing to achieve rates other than $R = \frac{k}{n}$. We define a puncturing pattern p , parse the n code sequence into blocks of the length of p and drop the bits corresponding to 0 in p .
- e.g. $p = [1, 1, 1, 0]$ $x = [1, 0, 1, 1, 1, 0, 1, 1]$ → $x' = [1, 0, 1, 1, 1, 0]$
- A rate $\frac{1}{n}$ encoder w/ this p has an overall rate of $R = \frac{1}{n} \times \frac{4}{3}$ 4 into bits 3 transmitted bits
- All known decoders can deal w/ puncturing by considering them as erased bits in the original non-punctured code

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D-transform

- The D transform maps sequences to power series. For a sequence $\{a_k\}$.

$$A(D) = \sum_{k=0}^{\infty} a_k D^k$$

For vectors of length l , $a_k = 0$ for $k \geq l \rightarrow (l-1)$ -degree polynomial.

- The D transform has the convolution property (maps convolution to multiplication).

$$\{x_k\} = \{u_k\} * \{c_k\} \xrightarrow{\text{D transform}} X(D) = U(D) C(D)$$

- Note the o/p of a convolutional code $\{x_k\}$ can be written as a convolution b/w the i/p $\{u_k\}$ and the connection vector c

$$\rightarrow \text{e.g. } x_k^{(2)} = u_k + u_{k-1} + u_{k-2} \Leftrightarrow \{x_k^{(2)}\} = \{u_k\} * \{c_k\} \text{ where } \{c_k\} = [1, 1, 1]$$

$$\therefore X^{(2)}(D) = U(D) C(D), \text{ where } C(D) = 1 + D + D^2.$$

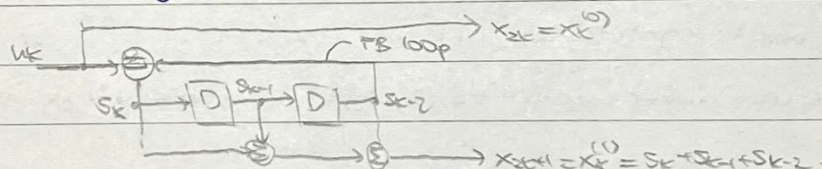
Systematic and recursive encoders.

- A systematic encoder has $x_k^{(0)} = u_k \rightarrow$ i/p sequence is every n th bit of o/p sequence.

(we can easily figure out the i/p sequence from the o/p sequence).

- A recursive encoder has a FB loop

$\rightarrow$ e.g. consider the following encoder.



$$x_k^{(1)} = s_k + s_{k-1} + s_{k-2} \rightarrow X^{(1)}(D) = (1 + D + D^2) S(D).$$

$$s_k = u_k + s_{k-2} \rightarrow u_k = s_k - s_{k-2} = s_k + s_{k+2} \rightarrow U(D) = (1 + D^2) S(D)$$

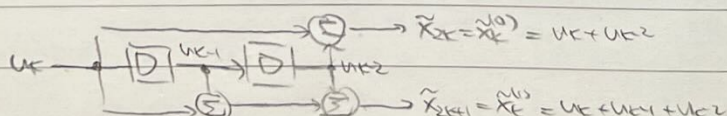
$$\therefore X^{(1)}(D) = \frac{1 + D + D^2}{1 + D^2} U(D)$$

Code vs encoder.

- We can have different encoders (mapping of i/p to o/p) give the same codes (same distance properties).

$\rightarrow$ codes are not unique

$\rightarrow$ e.g. consider the $(5,7)_8$ encoder.



$$\tilde{x}_k^{(0)} = u_k + u_{k-2} \rightarrow \tilde{X}^{(0)}(D) = (1 + D^2) U(D)$$

$$\tilde{x}_k^{(1)} = u_k + u_{k-1} + u_{k-2} \rightarrow \tilde{X}^{(1)}(D) = (1 + D + D^2) U(D)$$

Note that the o/p of the $(5,7)_8$ encoder $\tilde{x}_k^{(0)}, \tilde{x}_k^{(1)}$ are related to the o/p of the recursive encoder $x_k^{(0)}, x_k^{(1)}$

$$x_k^{(0)}(D) = \frac{\tilde{x}_k^{(0)}(D)}{1 + D^2}$$

$$x_k^{(1)}(D) = \frac{\tilde{x}_k^{(1)}(D)}{1 + D^2}$$

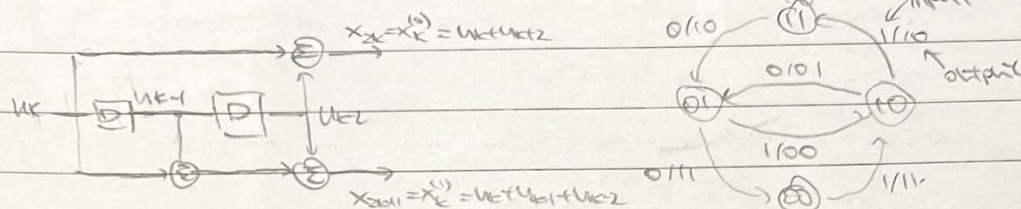
By dividing all o/p sequences by $(1 + D^2)$ (note 1 code), we do not change the set of possible sequences.

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State diagram

- A convolutional encoder is a finite state machine - we can draw a state diagram (edges labelled w/ i/p/o/p)

↳ e.g. Consider the (5,7)₈ encoder. We define the state $s = [u_{k-1} u_{k-2}]$



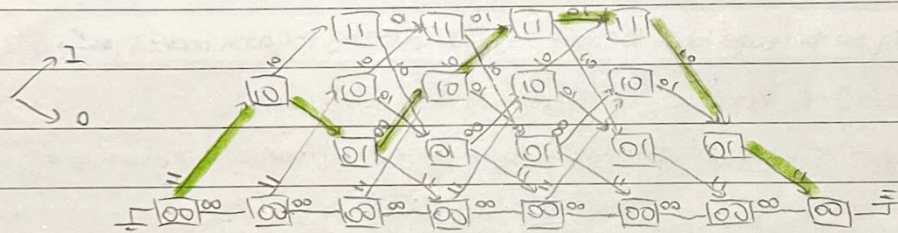
↳ we can ignore redundant delays → shift everything $u_{k-1} \rightarrow u_k$ etc

Trellis diagram

- The trellis diagram is an "unrolled" state diagram. States are in the y-dir; time is in the x-dir.

Branches are labelled w/ the o/p (i/p not labelled → always branch going up → 1 down → 0)

↳ e.g. Consider the (5,7)₈ encoder.



- The start node is the root and the end node is the leaf. The trellis diagram consists of an initial build-up until all the states are attained, followed by the repetition of a regular "trellis module".

- Every path in the trellis corresponds to a codeword.

↳ The highlighted path gives the codeword $x = [11 01 00 10 01 10 11]$ which corresponds to the input $y = [\text{information bits } 1011100 \text{ termination bits } 00]$

- A terminated convolutional code returns the encoder to the terminated state by inserting z symbols. (Insert two "0"s for forward encoders, state dependent for recursive encoders)

- Termination affects the rate of the code as we have dummy termination symbols

↳ The rate of the code is $R = \frac{5}{14}$, which is lower than $R = \frac{1}{2}$ of the convolutional code.

- In practice, termination only occurs after 100+ trellis modules → barely affect the overall rate.

* Note that termination is not always used - depends on application.

Convolutional code vs linear block code

sum of i/p sequences gives sum of o/p sequences.

- All operations in a convolutional code is linear → convolutional code is a linear code

- A terminated convolutional code is a (linear) block code → properties of LBCs apply to convolutional codes (distance-weight symmetry, correct up to t errors, $t \leq \lfloor \frac{d_{min}-1}{2} \rfloor$)

↳ also for convolutional codes

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Trellis based algorithms

Optimal decoding.

- We have bit-optimal or block-optimal decoding.

bit-optimal: $\hat{x} = \underset{x \in \{0,1\}^n}{\operatorname{argmax}} P(X_2=x|Y)$ where $P(X_2=x|Y) = \sum_{\substack{\text{sum over } 2^k \text{ codewords} \\ \text{over all } x \in \{0,1\}^n}} P(x|Y)$

block-optimal: $\hat{x} = \underset{x}{\operatorname{argmax}} P(x|Y)$
 $\leftarrow \text{max over } 2^k \text{ codewords}$

$\rightarrow$ we want to efficiently compute sums / max over 2^k terms.

Viterbi algorithm (block-optimal decoding) for the BSC.

- For a BSC w/ $p < 0.5$ (w/ just swap the 0/1s)

$$p(Y|X) = p^{d(Y,X)} (1-p)^{n-d(Y,X)} = (1-p)^n \left(\frac{p}{1-p}\right)^{d(Y,X)}$$

$$\therefore \underset{x}{\operatorname{argmax}} P(Y|x) = \underset{x}{\operatorname{argmin}} d(Y,x)$$

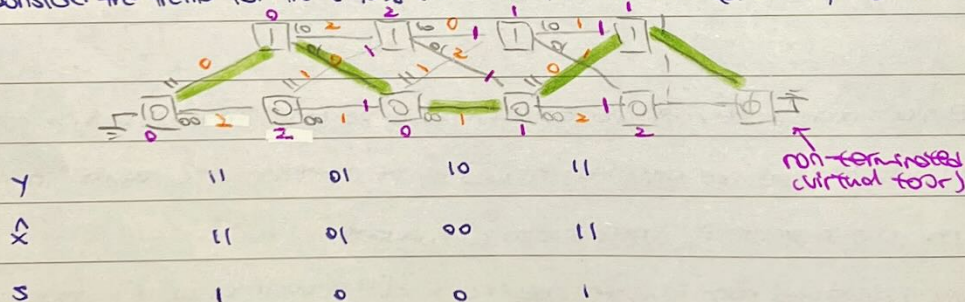
i.e. max likelihood decoder for BSC is a min. Hamming distance decoder.

- The overall min (over all codewords x) can be found using a trellis.

set edge γ_{ij} to the Hamming distance $d(Y,x)$, for each node j , compute $\alpha_j = \min(\gamma_{ij} + \alpha_i)$
 $\leftarrow \text{over all branches ending at node } j$

Initialise w/ $\alpha_{\text{root}} = 0 \rightarrow \alpha_{\text{root}}$ gives the overall min.

- eg: consider the trellis for the $(2,3)_3$ encoder. The received sequence is $Y = [11011]$



Viterbi algorithm (block-optimal decoding) for the BIAWGN

- For a BIAWGN x_k are interpreted as +1 (for 0) and -1 (for 1)

$$p(Y|X) = \prod_{k=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_k - x_k)^2}{2\sigma^2}\right)$$

$$\therefore \underset{x}{\operatorname{argmax}} P(Y|x) = \underset{x}{\operatorname{argmin}} \sum_{k=1}^n (Y_k - x_k)^2$$

i.e. max likelihood decoder for BIAWGN is a min. Euclidean distance decoder

- The overall min. (over all codewords x) can be found using a trellis.

set edge γ_{ij} to the Euclidean distance b/w Y, x . For each node j , compute $\alpha_j = \min(\gamma_{ij} + \alpha_i)$
 $\leftarrow \text{over all branches ending at node } j$

Initialise w/ $\alpha_{\text{root}} = 0 \rightarrow \alpha_{\text{root}}$ gives the overall min.

* Same as for BSC, but the edge labels γ_{ij} are different.

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BCJR algorithm (bit-optimal decoding) for the BSC.

- The partial sum can be found using a trellis.

Set edge γ_{ij} to the probability $p(y|x)$. For each node j , compute $\alpha_j = \sum_i \gamma_{ij} \alpha_i$, $\beta_j = \sum_i \gamma_{ij} \beta_i$.
 Initialise w/ $\alpha_{start} = \beta_{end} = 1 \rightarrow \alpha_i \gamma_{ij} \beta_j$ is the sum over all paths passing through that branch.
 (We get the partial sum by adding all the relevant branches).

Performance of Viterbi vs BCJR decoder.

- For any decoder, the bit error freq. is lower than the block error freq. since any single bit error in a block would lead to a block error.

- The Viterbi/BCJR decoder is optimal for block/bit decoding, so it would have a lower error freq.

$\rightarrow$ Given $f_1 < f_2 < f_3 < f_4$, f_1 : bit, BCJR; f_2 : bit, Viterbi; f_3 : block, Viterbi; f_4 : block, BCJR.

Properties of convolutional codes

Catastrophic encoders

- A convolutional encoder is catastrophic if there exists two (arbitrarily long) code sequences which only differ in a small no. of bits, but the corresponding two i/p sequences differ in an arbitrarily large no. of bits.

- With a catastrophic encoder, when one of these code sequences is transmitted, a small no. of channel errors can lead to an arbitrary large no. of errors in the i/p sequence.

- eg: consider the (6,5)₈ encoder

$$u = 1111... \rightarrow x = 11 \underline{01} \underline{00} 00 00 00...$$

$$u' = 1000... \rightarrow x' = 11 \underline{10} \underline{01} 00 00 00...$$

suppose we want to encode $u \rightarrow$ transmit $x \rightarrow$ receive $x' \rightarrow$ decode u'

i.e. A finite no. of channel errors leads to infinitely many decoding errors $\rightarrow$ catastrophic.

- Alternatively, consider $u' = 0000...$, $x' = 0002...$, then an encoder is catastrophic when an infinite weight i/p sequence results in a (small) finite weight o/p sequence x .

- Suppose a rate $R=1/n$ encoder has generators $\{c_1, c_2, \dots, c_n\}$. If the greatest common divisor of the polynomials $\{c_1(D), c_2(D), \dots, c_n(D)\}$ is of the form D^l for some integer $l \geq 0$, then the encoder is not catastrophic. (i.e. the polynomials have no common divisor that gives a finite weight o/p $\rightarrow$ not catastrophic).

1. Rate $R=1$ encoders are always catastrophic as we can set $U(D) = \frac{1}{C(D)}$

- eg: Consider the $R=1/2$ encoder w/ generators $C_1(D) = 1+D$, $C_2(D) = 1+D^2 = (1+D)^2$

choose $U(D) = \frac{1}{1+D}$ (i.e. $u = 1, 1, 1, \dots$), $X^{(1)}(D) = C_1(D)U(D) = 1$

$X^{(2)}(D) = C_2(D)U(D) = 1+D$

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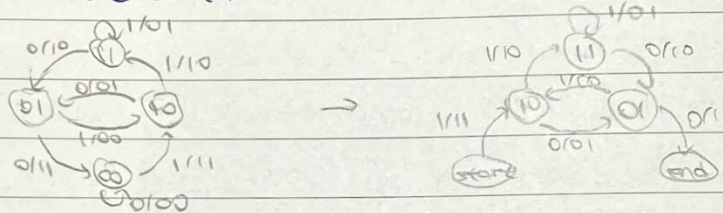
Detours in trellises.

- consider we transmit the all-zero codeword. we consider the detours (caused by channel noise) from the all-zero path. (sufficient for measuring decoder performance since the code is linear and the channel is BSCMC).

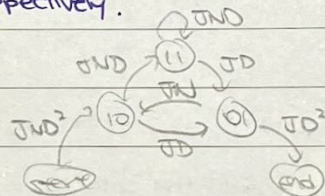
- A measure of decoder performance is the free distance d_{free} , the defn of min. v.p weight.

- We can find d_{free} by determining the extended transfer function of the code $T(J, N, D)$.

① Convert the state diagram into a transfer diagram by splitting the zero state into a "start" and "end" state.



② Label each branch w/ $JN^w D^{w_0}$, where w_1 and w_0 are the weights of the v.p bit and o.p bits along the branch respectively.



③ Find the extended transfer function $T(J, N, D)$ via system of eqns or Mason's gain formula.

System of eqns Define a variable for every state and set up a system of eqns

$$\begin{cases} X_{10} = JND^2 X_{\text{start}} + JNX_{01} \\ X_{11} = JNDX_{11} + JNDX_{10} \\ X_{01} = JDX_{10} + JDX_{11} \\ X_{\text{end}} = JD^2 X_{01} \end{cases} \rightarrow T(J, N, D) = \frac{J^3 ND^5}{1 - (1+J)JND}$$

Solve for $T(J, N, D) = \frac{X_{\text{end}}}{X_{\text{start}}}$ analytically.

Mason's gain formula $T(J, N, D) = \frac{X_{\text{end}}}{X_{\text{start}}} = \frac{\sum_k G_k \Delta_k}{\Delta}$

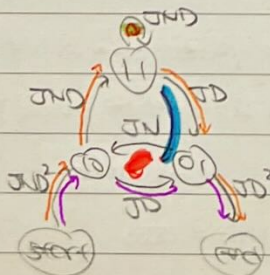
where $\Delta = 1 - \sum_i L_i + \sum_{i,j} L_i L_j - \sum_{i,j,k} L_i L_j L_k + \dots$ is the graph determinant

G_k is the gain of the k -th forward path

L_i is the gain of the i -th loop

L_i is the product of non-intersecting loops

Δ_k is the graph determinant w/ loops intersecting w/ left forward path removed



$$\Delta = 1 - JND - J^3 ND^3 - J^2 ND + (JND)(J^2 ND)$$

$$G_1 = J^3 ND^5, \Delta_1 = 1 - JND \quad \text{forward path 1}$$

$$G_2 = J^4 ND^6, \Delta_2 = 1 - 0 \quad \text{forward path 2}$$

$$\Rightarrow T(J, N, D) = \frac{G_1 \Delta_1 + G_2 \Delta_2}{\Delta} = \frac{J^3 ND^5 (1 - JND) + J^4 ND^6}{1 - JND - J^3 ND^3 - J^2 ND + (JND)(J^2 ND)} = \frac{J^3 ND^5}{1 - (1+J)JND}$$

- Expand $T(J, N, D)$ into a power series, $T(J, N, D) = J^3 ND^5 (1 + \alpha + \alpha^2 + \dots)$ where $\alpha = (1+J)JND$. Arranging in increasing powers of D , the first term gives d_{free} ($d_{\text{free}} = 5$)

Signal space concepts

Mapping bits to waveforms.

- Suppose we want to transmit n bits in T seconds using a waveform $x(t)$, $t \in [0, T]$.
- We choose a collection of $M=2^n$ waveforms $\{s_1(t), \dots, s_M(t)\}$ and map each n -bit sequence to one of these waveforms.
- To systematically choose the $M=2^n$ finite-energy waveforms, we can consider a set of basis functions in the signal space.

Vector spaces.

- A vector space V is a set of elements (vectors) that is closed under addition and scalar multiplication. i.e. if $\underline{v}_1, \underline{v}_2 \in V$, then $a\underline{v}_1 + b\underline{v}_2 \in V$ for any scalars a, b .
(The scalars are often real/complex, but could also be chosen over some other field).
- A set of linearly independent vectors, say $\{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k\}$ form a basis of the vector space V , if every $\underline{v} \in V$ can be expressed as a linear combination of the form ↑ not unique

$$\underline{v} = a_1 \underline{v}_1 + a_2 \underline{v}_2 + \dots + a_k \underline{v}_k$$

If the no. of vectors in the basis is a finite no., the vector space has dimension k .

- If we denote the inner product between two vectors $\underline{v}_1, \underline{v}_2$, as $\langle \underline{v}_1, \underline{v}_2 \rangle$, then

(i) $\underline{v}_1$ and $\underline{v}_2$ are orthogonal if $\langle \underline{v}_1, \underline{v}_2 \rangle = 0$

(ii) the L_2 norm of a vector is defined as $\|\underline{v}\|_2 = \sqrt{\langle \underline{v}, \underline{v} \rangle}$.

vector space of signals — signal space.

- Let L_2 be the set of complex-valued signals (functions) $x(t)$ w/ finite energy, i.e. $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$.

It can be shown that L_2 is a vector space.

- We define the inner product in this space as follows. For $x(t), y(t) \in L_2$,

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y^*(t) dt$$

so $x(t), y(t)$ are orthogonal if $\int_{-\infty}^{\infty} x(t) y^*(t) dt = 0$ and $\|\underline{x}\|_2 = \left[\int_{-\infty}^{\infty} |x(t)|^2 dt \right]^{1/2}$ (square root of energy).

- For any vector space $L \subset L_2$, the set of functions $\{f_i(t), i=1, 2, \dots\}$ is an orthonormal basis for L if

(i) Every $x(t) \in L$ can be expressed as $x(t) = \sum_i x_i f_i(t)$ where x_i are scalars.

(ii) the functions $f_i(t), i=1, 2, \dots$ are orthonormal, i.e. $\langle f_l, f_m \rangle = \int_{-\infty}^{\infty} f_l(t) f_m^*(t) dt = \begin{cases} 1 & \text{if } l=m \\ 0 & \text{if } l \neq m \end{cases}$

If the two conditions above are satisfied, the orthonormal basis $\{f_i(t), i=1, 2, \dots\}$ is said to span L .

- The no. of elements (functions) in the basis is the dimension of L .

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Gram-Schmidt procedure in the signal space.

- The Gram-Schmidt procedure gives a systematic way to find an orthonormal basis given a set of signals.

- Given functions $\{x_1(t), x_2(t), \dots, x_n(t)\}$, we can find an orthonormal basis $\{f_1(t), f_2(t), \dots\}$ as follows:

$$\hookrightarrow 1) \text{ Let } f_1(t) = \frac{x_1(t)}{\|x_1(t)\|}$$

$$\hookrightarrow 2) \text{ Let } g_2(t) = x_2(t) - \langle x_2, f_1 \rangle f_1(t)$$

[component of x_2 orthogonal to f_1]

$$\text{Then } f_2(t) = \frac{g_2(t)}{\|g_2(t)\|}$$

$$\hookrightarrow 3) \text{ Let } g_3(t) = x_3(t) - \langle x_3, f_1 \rangle f_1(t) - \langle x_3, f_2 \rangle f_2(t) \quad [\text{component of } x_3 \text{ orthogonal to } f_1, f_2]$$

$$\text{Then } f_3(t) = \frac{g_3(t)}{\|g_3(t)\|} \quad \text{etc.}$$

If the dimension of the space is k , only $f_1, \dots, f_k$ will be non-zero.

Functions in signal space and vectors in Euclidean space.

- Consider any function $x(t) \in L$ expressed in terms of the orthonormal basis $\{f_i(t)\}$, $x(t) = \sum_i x_i f_i(t)$

Each coefficient x_j can be calculated as $x_j = \langle x, f_j \rangle$

$$\langle x, f_j \rangle = \int_{-\infty}^{\infty} x(t) f_j(t) dt = \int_{-\infty}^{\infty} \left(\sum_i x_i f_i(t) \right) f_j(t) dt = \sum_i x_i \int_{-\infty}^{\infty} f_i(t) f_j(t) dt = x_j$$

- The coefficients $x_1, x_2, \dots$ are the projection coefficients and $x_1 f_1(t), x_2 f_2(t), \dots$ are the projections of the signal $x(t)$ along $f_1(t), f_2(t), \dots$ respectively.

- The inner product between $x(t), y(t)$ is $\langle x, y \rangle = \sum_i x_i y_i^*$

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y(t)^* dt = \int_{-\infty}^{\infty} \left(\sum_i x_i f_i(t) \right) \left(\sum_j y_j f_j(t)^* \right) dt = \sum_{i,j} x_i y_j^* \int_{-\infty}^{\infty} f_i(t) f_j(t)^* dt = \sum_i x_i y_i^*$$

The energy of $x(t)$ can therefore be written as $\int_{-\infty}^{\infty} |x(t)|^2 dt = \sum_i |x_i|^2$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \langle x, x \rangle = \sum_i |x_i|^2$$

- If we fix an orthonormal basis for L , we can treat functions in L just like vectors in Euclidean space

$$x(t) \mapsto (x_1, x_2, \dots, x_k)$$

where k is the no. of elements in the orthonormal basis (dimension of signal space).

→ easy to calculate inner products / energies as we effectively converted continuous-time operations (integrals over t) into discrete-time operations (summations).

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Modelling a channel

Modelling a channel

- channels are often modelled as LTI systems w/ additive noise.

With such a model, the channel output $y(t)$ generated from input $x(t)$ is given by

$$y(t) = h(t) * x(t) + n(t)$$

and in the freq. domain,

$$Y(f) = H(f)X(f) + N(f)$$

- In the special case where $|H(f)|$ is const. throughout the BW of i/p $x(t)$ and we have a const. time delay τ .

(i.e. $\angle H(f) = -2\pi f\tau$), then we can use the additive noise model

ej: telephone cables for voice only

$$y(t) = x(t) + n(t)$$

$$\rightarrow |H(f)| = c, \angle H(f) = -2\pi f\tau \rightarrow H(f) = ce^{-j2\pi f\tau} \text{ so } Y(f) = ce^{-j2\pi f\tau} X(f) + N(f) \rightarrow y(t) = cx(t-\tau) + n(t)$$

We can compensate for the const. channel gain c / delay τ at the receiver $\rightarrow y(t) = x(t) + n(t)$.

- We can no longer use the additive noise model if

$\rightarrow$ 1) The i/p signal $x(t)$ has a much larger BW $\rightarrow |H(f)|$ no longer const. throughout its BW

$\rightarrow$ 2) There are multiple signal paths from Tx to Rx (mobile wireless comms - passband).

delays aren't const if Tx/Rx are mobile.

Say we have L signal paths w/ delays $\tau_1, \dots, \tau_L$ and attenuations $\alpha_1, \dots, \alpha_L$, the received signal is.

$$y(t) = \alpha_1 x(t-\tau_1) + \dots + \alpha_L x(t-\tau_L) + n(t) = x(t) * h(t) + n(t)$$

where $h(t) = \alpha_1 \delta(t-\tau_1) + \dots + \alpha_L \delta(t-\tau_L) \rightarrow |H(f)|$ no longer const. throughout the BW of i/p $x(t)$.

Noise model

- In general, $y(t) = h(t) * x(t) + n(t)$, where $n(t)$ is thermal noise at the Rx:

$\rightarrow$ thermal noise is the noise generated by the thermal agitation of e^- inside an electrical conductor

$\rightarrow$ there is thermal noise regardless of the applied voltage / Rx.

- consider modelling $n(t)$ as a Gaussian white noise process. If we operate in a freq. band where $|H(f)|$ is flat,

then the effective channel is an additive Gaussian white noise (AWGN) channel, $y(t) = x(t) + n(t)$, where

$$E[n(t)] = 0 \quad \text{and} \quad E[n(t)n(t+\tau)] = \frac{N_0}{2} \delta(\tau) \quad \text{for all } t, \tau.$$

- Note this implies that the PSD $S_n(f)$ is given by

$$S_n(f) = \frac{N_0}{2} \quad -\infty < f < \infty.$$

Although the power $E[\text{Integr.}] = \int_{-\infty}^{\infty} S_n(f) df$ is infinite, this is not a problem in practice as the transmitted

signal $x(t)$ is bandlimited, say $[B, B] \rightarrow$ received waveform $y(t)$ passed through a filter that rejects

all freq. components outside a band, say $[B, B] \rightarrow$ we effectively have $S_n(f) = \frac{N_0}{2} \quad -B < f < B$.

* We still take $S_n(f) = \frac{N_0}{2}$ for all f for mathematical convenience.

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From waveform detection to vector detection

- The signal set consists of M waveforms $\{s_1(t), \dots, s_M(t)\}$, and we transmit one of M messages using this signal set - to transmit message i , we set $x(t) = s_i(t) \rightarrow$ we want to detect $x(t) = s_i(t)$ from $y(t)$.

- Suppose that $\{\phi_1(t), \dots, \phi_K(t)\}$ form an orthonormal basis for the signal set (KSM). Then each signal $s_i(t)$ is

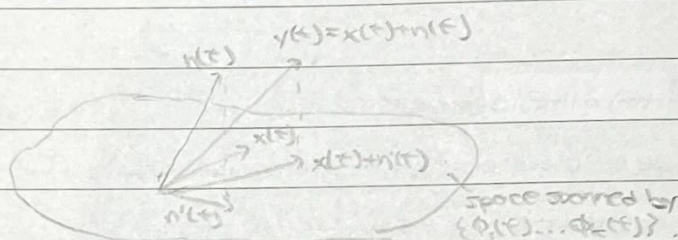
$$s_i(t) = s_{i,1}\phi_1(t) + \dots + s_{i,K}\phi_K(t)$$

where the j th projection coefficient for $s_i(t)$ is

$$s_{i,j} = \langle s_i, \phi_j \rangle = \int s_i(t) \phi_j(t) dt$$

$\rightarrow$ each signal is equivalent to a K -dimensional vector. $s_i(t) \leftrightarrow \underline{s}_i = [s_{i,1}, \dots, s_{i,K}]$

- consider a geometric representation of the signals



the transmitted signal $x(t)$ is one of $\{s_1(t), \dots, s_M(t)\} \rightarrow$ lie in the space spanned by $\{\phi_1(t), \dots, \phi_K(t)\}$.

In general the noise waveform $n(t)$ will not lie in the signal space spanned by $\{\phi_1(t), \dots, \phi_K(t)\}$.

$\rightarrow$ Only the component of the noise in the signal space matters for detecting $\rightarrow$ Rx can project the received waveform onto the space spanned by $\{\phi_1(t), \dots, \phi_K(t)\}$ w/o losing any useful information.

- Let $\underline{x} = [x_1, \dots, x_K]$, $\underline{n} = [n_1, \dots, n_K]$ and $\underline{y} = [y_1, \dots, y_K]$ be the projection coeff. vectors of $x(t)$, $n(t)$ and $y(t)$ respectively.

$$x_j = \langle x, \phi_j \rangle$$

$$n_j = \langle n, \phi_j \rangle$$

$$y_j = \langle y, \phi_j \rangle$$

Note that we can write $\underline{y} = \underline{x} + \underline{n}$, since y_j is given by

$$y_j = \langle y, \phi_j \rangle = \langle x + n, \phi_j \rangle = \langle x, \phi_j \rangle + \langle n, \phi_j \rangle = x_j + n_j$$

If the transmitted signal is $x(t) = s_i(t)$, then $\underline{x} = \underline{s}_i$.

$\rightarrow$ we have reduced the waveform detection problem into a simpler vector detection problem.
 detect $x(t) = s_i(t)$ from $y(t)$
 detect $\underline{x} = \underline{s}_i$ from $\underline{y}$

The distribution of the projected noise vector $\underline{n}$.

- Let $\{\phi_k(t)\}_{k=1}^K$ be any orthonormal set of functions and $n(t)$ be a Gaussian noise white noise process, i.e.

$$E[n(t)] = 0 \quad \text{and} \quad E[n(t)n(t')] = \frac{N_0}{2} \delta(t-t') \quad \text{for all } t, t'.$$

For all $m \in \mathbb{Z}$, define $n_m = \langle n, \phi_m \rangle$, then $\{n_m\}_{m=1}^K$ are iid Gaussian w/ zero mean and variance $\frac{N_0}{2}$.

- n_m is a linear combination of jointly Gaussian RVs $\{n(t), t \in \mathbb{R}\} \rightarrow$ RVs $\{n_m\}_{m=1}^K$ are jointly Gaussian.

$\hookrightarrow$ Mean $E[n_m]$. For each m ,

$$E[n_m] = E[\langle n, \phi_m \rangle] = E\left[\int_{-\infty}^{\infty} n(t) \phi_m(t) dt\right] = \int_{-\infty}^{\infty} E[n(t)] \phi_m(t) dt = 0$$

$\hookrightarrow$ covariance $E[n_m n_l^*]$. For each pair (m, l) .

$$E[n_m n_l^*] = E[\langle n, \phi_m \rangle \langle n, \phi_l \rangle^*] = E\left[\left(\int_{-\infty}^{\infty} n(t) \phi_m(t) dt\right) \left(\int_{-\infty}^{\infty} n(s) \phi_l^*(s) ds\right)^*\right] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E[n(t) n(s)^*] \phi_m(t) \phi_l^*(s) dt ds$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{N_0}{2} \delta(t-s) \phi_m(t) \phi_l^*(s) ds = \frac{N_0}{2} \int_{-\infty}^{\infty} \phi_m(t) \phi_l^*(t) dt = \begin{cases} N_0/2 & \text{if } m=l \\ 0 & \text{if } m \neq l \end{cases}$$

$\rightarrow \{n_m\}_{m=1}^K$ are iid Gaussian w/ zero mean and variance $N_0/2$

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Optimal detection

- The set of signal vectors is $S = \{\underline{s}_1, \dots, \underline{s}_M\}$, and each signal vector in S has some prior probability of being transmitted $P(\underline{x} = \underline{s}_i)$.
- We want the receiver to decide the vector $\hat{\underline{x}}$, s.t. it min. the probability of detection error $P_e = P(\hat{\underline{x}} \neq \underline{x})$.
- In general (not only AWGN), the optimal detection rule that min. P_e is the MAP rule.

$$\hat{\underline{x}}^{MAP} = \underset{\underline{s}_i \in S}{\operatorname{argmax}} \overset{\text{posterior}}{P(\underline{x} = \underline{s}_i | \underline{r})} = \underset{\underline{s}_i \in S}{\operatorname{argmax}} \overset{\text{prior}}{P(\underline{x} = \underline{s}_i)} \overset{\text{likelihood}}{P(\underline{r} | \underline{x} = \underline{s}_i)}$$

↳ The MAP rule is optimal. Consider $\hat{\underline{x}} = g(\underline{r})$ to be the decoded symbol using any other decoding rule g .

$$P(\hat{\underline{x}} = \underline{x}) = \int_{\underline{r}} P(\hat{\underline{x}}(\underline{r}) = \underline{x} | \underline{r}) P(\underline{r}) d\underline{r} \leq \int_{\underline{r}} \max_{\underline{s}_i \in S} P(\underline{s}_i = \underline{x} | \underline{r}) P(\underline{r}) d\underline{r} = \int_{\underline{r}} P(\hat{\underline{x}}(\underline{r}) = \underline{x} | \underline{r}) P(\underline{r}) d\underline{r} = P(\hat{\underline{x}} = \underline{x})$$

i.e. any other decoding rule cannot do better than the MAP rule $\rightarrow$ it is optimal.

- If the prior distribution on the signal vectors is uniform, i.e. $P(\underline{x} = \underline{s}_i)$ is const. for all symbols $\underline{s}_i \in S$, then

$$\hat{\underline{x}}^{ML} = \underset{\underline{s}_i \in S}{\operatorname{argmax}} \overset{\text{likelihood}}{P(\underline{r} | \underline{x} = \underline{s}_i)}$$

i.e. MAP reduces to ML when the prior distribution is uniform.

Optimal detection for AWGN

- For AWGN $\underline{n}(t)$, $\underline{n} = [n_1, \dots, n_K]$ is a Gaussian random vector w/ n_j iid $\mathcal{N}(0, N_0/2)$

Since $\underline{r}_j = \underline{x}_j + \underline{n}_j$, given that $\underline{x}_j = \underline{s}_{i,j}$,

$$P(\underline{r}_j | \underline{x}_j = \underline{s}_{i,j}) = \mathcal{N}(\underline{r}_j | \underline{s}_{i,j}, \frac{N_0}{2}) = \frac{1}{(\pi N_0)^{K/2}} \exp\left(-\frac{(\underline{r}_j - \underline{s}_{i,j})^T}{N_0}\right)$$

therefore the likelihood $P(\underline{r} | \underline{x} = \underline{s}_i)$ is given by

$$P(\underline{r} | \underline{x} = \underline{s}_i) = \prod_{j=1}^K \frac{1}{(\pi N_0)^{K/2}} \exp\left(-\frac{(\underline{r}_j - \underline{s}_{i,j})^T}{N_0}\right) = \frac{1}{(\pi N_0)^{K/2}} \exp\left(-\frac{\|\underline{r} - \underline{s}_i\|^2}{N_0}\right)$$

where $\|\underline{r} - \underline{s}_i\|^2 = \sum_j (\underline{r}_j - \underline{s}_{i,j})^2$ is the squared distance between $\underline{r}$ and $\underline{s}_i$.

- If the prior distribution over the signal vectors is uniform, the optimal detection rule is the ML rule.

$$\hat{\underline{x}}^{ML} = \underset{\underline{s}_i \in S}{\operatorname{argmax}} P(\underline{r} | \underline{x} = \underline{s}_i) = \underset{\underline{s}_i \in S}{\operatorname{argmin}} \|\underline{r} - \underline{s}_i\|^2$$

i.e. the detector chooses the signal closest to the received vector $\rightarrow$ min. distance decoding.

- Suppose that the prior probabilities of the signal vectors are $P(\underline{x} = \underline{s}_i) = p_i$, the posterior $P(\underline{x} = \underline{s}_i | \underline{r})$ is given by

$$P(\underline{x} = \underline{s}_i | \underline{r}) \propto P(\underline{x} = \underline{s}_i) P(\underline{r} | \underline{x} = \underline{s}_i) = p_i \frac{1}{(\pi N_0)^{K/2}} \exp\left(-\frac{\|\underline{r} - \underline{s}_i\|^2}{N_0}\right) = \frac{1}{(\pi N_0)^{K/2}} \exp\left(-\ln p_i + \frac{\|\underline{r} - \underline{s}_i\|^2}{N_0}\right)$$

- In the general case where the prior distribution over the signal vectors is $P(\underline{x} = \underline{s}_i) = p_i$, the optimal detection is the MAP rule.

$$\hat{\underline{x}}^{MAP} = \underset{\underline{s}_i \in S}{\operatorname{argmax}} P(\underline{x} = \underline{s}_i | \underline{r}) = \underset{\underline{s}_i \in S}{\operatorname{argmin}} \ln p_i + \frac{\|\underline{r} - \underline{s}_i\|^2}{N_0}$$

i.e. the detector trades-off between the prior probability and distance to the received vector.

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Probability of detection error P_e

- In general, the probability of detection error P_e is given by

$$P_e = P(\hat{x} \neq x) = \sum_i P(x = s_i) P(\hat{x} \neq x | x = s_i)$$

* If there is symmetry in the signals and their prior, we can assume $x = s_1$ was transmitted w/o loss of generality

$$P_e = P(\hat{x} \neq x | x = s_1)$$

- We can express P_e in terms of the Q-function by making n the subject in the probability expression, then dividing by the std. $\sqrt{N_0/2}$ to get the standard normal distribution.

$$P\left(\frac{n}{\sqrt{N_0/2}} < -p\right) = P\left(\frac{n}{\sqrt{N_0/2}} > p\right) = Q(p) \quad \text{where } Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}u^2) du$$

$$Q(x) = P(X > x) \text{ for } X \sim N(0,1)$$

- This is useful as we can use the following bound to express P_e in terms of the SNR

$$Q(x) \leq \frac{1}{2} e^{-x^2/2} \quad x \geq 0$$

* If P_e is in the form $P_e = aQ(x) - bQ(x)^2$, then we can use the upper bound $P_e \leq aQ(x)$.

The union bound.

- often we cannot compute P_e exactly, but we can accurately bound it to the union bound. Assuming

that each signal is equally likely, then P_e is given by

$$P_e = \frac{1}{M} \sum_{i=1}^M P(\text{error} | x = s_i) = \frac{1}{M} \sum_{i=1}^M P\left(\bigcup_{j \neq i} \{\hat{x}(r) = s_j\} | x = s_i\right) \leq \frac{1}{M} \sum_{i=1}^M \sum_{j \neq i} P(\hat{x}(r) = s_j | x = s_i)$$

- For AWGN $n(t)$, the union bound becomes

$$P_e \leq \frac{1}{M} \sum_{i=1}^M \sum_{j \neq i} Q\left(\sqrt{\frac{\|s_i - s_j\|^2}{2N_0}}\right)$$

$$\hookrightarrow P(\hat{x}(r) = s_j | x = s_i) = P(\|r - s_j\|^2 < \|r - s_i\|^2 | x = s_i) = P(\|s_i - s_j + n\|^2 \leq \|n\|^2 | x = s_i)$$

$$= P(\|s_i - s_j\|^2 + \|n\|^2 + 2 \sum_{k=1}^K (s_{i,k} - s_{j,k}) n_k \leq \|n\|^2 | x = s_i) = P(Z_{ij} \leq -\|s_i - s_j\|^2 | x = s_i)$$

where $Z_{ij} = 2 \sum_{k=1}^K (s_{i,k} - s_{j,k}) n_k$ is the weighted sum of Gaussian RVs $\rightarrow$ it is also Gaussian.

$$\text{mean } E[Z_{ij}] = E\left[2 \sum_{k=1}^K (s_{i,k} - s_{j,k}) n_k\right] = 2 \sum_{k=1}^K (s_{i,k} - s_{j,k}) E[n_k] = 0$$

$$\text{variance } E[Z_{ij}^2] = E\left[\left(2 \sum_{k=1}^K (s_{i,k} - s_{j,k}) n_k\right)^2\right] = 4 \sum_{k=1}^K \sum_{l=1}^K (s_{i,k} - s_{j,k}) (s_{i,l} - s_{j,l}) E[n_k n_l]$$

$$\approx 2N_0 \sum_{k=1}^K (s_{i,k} - s_{j,k})^2 = 2N_0 \|s_i - s_j\|^2$$

$$\rightarrow Z_{ij} \sim N(0, 2N_0 \|s_i - s_j\|^2) \rightarrow \frac{Z_{ij}}{\sqrt{2N_0 \|s_i - s_j\|^2}} \sim N(0, 1)$$

$$\therefore P(\hat{x}(r) = s_j | x = s_i) = P(Z_{ij} \leq -\|s_i - s_j\|^2 | x = s_i) = P\left(\frac{Z_{ij}}{\sqrt{2N_0 \|s_i - s_j\|^2}} \leq -\sqrt{\frac{\|s_i - s_j\|^2}{2N_0}} | x = s_i\right) = Q\left(\sqrt{\frac{\|s_i - s_j\|^2}{2N_0}}\right)$$

Baseband transmission using PAM.

Pulse amplitude modulation (PAM).

- The PAM modulation scheme has two basic components:

→ 1) Mapping bits to real no. → constellation symbols.

→ 2) Modulate the baseband waveform $p(t)$ using the constellation symbols

- The set of values the bits are mapped to is the constellation.

→ eg: $0 \rightarrow A, 1 \rightarrow -A$ $\{-A, A\}$ // $00 \rightarrow 3A, 01 \rightarrow A, 10 \rightarrow -A, 11 \rightarrow -3A$ $\{-3A, -A, A, 3A\}$.

- Once we fix a constellation, a sequence of bits can be uniquely mapped to constellation symbols

→ eg: 010110010 ; using $\{-A, A\}$ $-A, A, -A, A, A, A, -A, -A, A, -A$

using $\{3A, -A, A, -3A\}$ $-A, -A, 3A, -3A, A.$

- In a constellation w/ M symbols, each symbol represents $\log_2 M$ bits.

- The baseband waveform $p(t)$ used in PAM is the pulse shape

Common pulse shapes are the sinc pulse and the rectangular pulse:

$$p(t) = \frac{1}{T} \text{sinc}\left(\frac{t}{T}\right)$$

$$p(t) = \frac{1}{T} \text{rect}\left(\frac{t}{T}\right)$$

where T is the symbol time of the pulse.

- The sequence of constellation symbols $x_0, x_1, \dots$ is used to modulate a series of pulses to generate a baseband signal as follows:

$$x(t) = \sum_k x_k p(t - kT)$$

- Every T seconds, a new symbol is introduced by shifting the pulse and modulating its amplitude w/ the symbol → transmission rate is $1/T$ symbols/s or $\frac{\log_2 M}{T}$ bits/s

Desirable properties of the pulse shape $p(t)$

- The pulse shape $p(t)$ is chosen to satisfy the following important objectives:

→ $p(t)$ decays quickly in time — the effect of x_k should not start much before $t = kT$ or last much beyond $t = (k+1)T$

→ $p(t)$ should be approx. band limited. For a fixed sequence of symbols $\{x_k\}$, the spectrum of $x(t)$ is

$$X(f) = \mathcal{F}[x(t)] = \mathcal{F}\left[\sum_k x_k p(t - kT)\right] = P(f) \sum_k x_k e^{-j2\pi f kT}$$

∴ BW of $x(t) \propto$ BW of $p(t)$ → band-limited pulse is req. for band-limited PAM signal.

→ the retrieval of the information sequence from the noisy received waveform $y(t) = x(t) + n(t)$

should be simple and rel. reliable. In the absence of noise, the symbols $\{x_k\}_{k \in \mathbb{Z}}$ should be

recovered perfectly at the receiver

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Time delay vs Bandwidth tradeoff

- Ideally, we want both quick decay in time and approx. bandlimited $p(t)$. However, a faster decay in time means a larger BW,
- For the sinc pulse, $p(t) = \frac{1}{\sqrt{T}} \text{sinc}(\frac{t}{T})$, it's perfectly bandlimited to $BW = \frac{1}{2T}$ but decays slowly in time $|p(t)| \sim 1/t$
- For the rectangular pulse, $p(t) = \frac{1}{\sqrt{T}} \text{rect}(t/T)$, it decays slowly in freq, $|p(f)| \sim 1/f$ but time limited to the interval $[-T/2, T/2]$
- In practice, the root-raised cosine pulse is used. It has BW slightly larger than $\frac{1}{2T}$, but decays much quicker in time, $|p(t)| \sim 1/t^2$

Orthogonality of pulse shifts

- To achieve simple and reliable detection, we choose $p(t)$ s.t. the shifted pulses $\{p(t-mT)\}_{m \in \mathbb{Z}}$ form an orthonormal basis. For $k, m \in \mathbb{Z}$,

$$\langle p(t-kT), p(t-mT) \rangle = \int_{-\infty}^{\infty} p(t-kT) p(t-mT) dt = \begin{cases} 1 & \text{if } k=m \\ 0 & \text{if } k \neq m \end{cases}$$

- Denoting $\phi_k(t) = p(t-mT)$, we have the orthonormal basis $\{\phi_k(t)\}_{k \in \mathbb{Z}}$.

The PAM signal is $x(t) = \sum_n x_n \phi_n(t) \rightarrow x_n$ is the projection coeff. of $x(t)$ w/ $\phi_n(t)$.

- Let N_n and Y_n be the projection coeff. of $n(t)$ and $y(t)$ w/ $\phi_n(t)$ respectively.

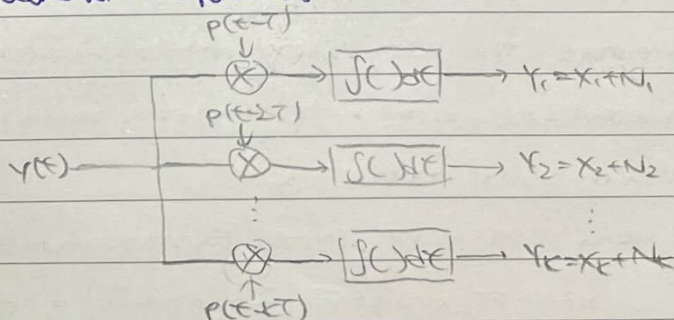
$$N_n = \langle n, \phi_n \rangle$$

$$Y_n = \langle y, \phi_n \rangle$$

Note that we can write $Y_n = X_n + N_n$ since

$$Y_n = \langle y, \phi_n \rangle = \langle x+n, \phi_n \rangle = \langle x, \phi_n \rangle + \langle n, \phi_n \rangle = X_n + N_n$$

- To detect symbols $x_1 \dots x_K$, the receiver can be implemented w/ a bank of K "correlators", which can compute the inner products w/ the K pulse shifts

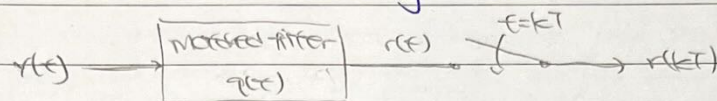


This implementation is not practical when we want to transmit a large amount of data (large K)

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Matched filter implementation

- The PAM receiver can be constructed by sampling the o/p of a matched filter



- Let the filter impulse response be $q(t) = p(t - T)$. Since $y(t) = \sum_n x_n p(t - nT) + n(t)$, the filter o/p $r(t)$ is

$$r(t) = y(t) * q(t) = x(t) * q(t) + n(t) * q(t) = \sum_n x_n \int_{-\infty}^{\infty} p(t - nT) p(t - T) d\tau + \int_{-\infty}^{\infty} n(\tau) p(t - T) d\tau$$

sampling at $t = kT$, $k = 1, 2, \dots$, we get $r(kT)$, given by

$$r(kT) = x_k + N_k$$

where $N_k = \int_{-\infty}^{\infty} n(\tau) p(t - kT) d\tau = \langle n, \phi_k \rangle$, the projection coeff. of $n(t)$ w/ $\phi_k(t)$.

- Note that if we denote $r(kT)$ by Y_k , we have an efficient (streaming) implementation of the optimal receiver which does not req. a separate correlator unit to detect each symbol.

Optimal detection for PAM

- We want to detect the symbol x_k from $Y_k = x_k + N_k$ for $k \in \mathbb{Z}$, where x_k is one of $c \in \mathcal{C}$.

where $N_1, N_2, \dots$ iid $N(0, N_0/2)$ since $n(t)$ is white Gaussian noise w/ PSD $N_0/2$.

- The optimal detection rule is the MAP rule. When $Y = y$,

$$\hat{x}^{\text{MAP}} = \underset{c \in \mathcal{C}}{\text{argmax}} P(x=c|Y) = \underset{c \in \mathcal{C}}{\text{argmax}} P(x=c) P(Y|x=c)$$

If all the symbols in the constellation are equally likely, i.e. $P(x=c)$ is const. for all $c \in \mathcal{C}$, then we have the ML rule

$$\hat{x}^{\text{ML}} = \underset{c \in \mathcal{C}}{\text{argmax}} P(Y|x=c)$$

- For PAM, $N \sim N(0, N_0/2)$ and $Y = x + N$, so for $x=c$, $Y \sim N(c, N_0/2)$. i.e.

$$p(Y|x=c) = \frac{1}{\sqrt{\pi N_0/2}} \exp\left(-\frac{(Y-c)^2}{N_0/2}\right)$$

The ML rule is therefore

$$\hat{x}^{\text{ML}} = \underset{c \in \mathcal{C}}{\text{argmin}} (Y-c)^2$$

Probability of detection error P_e and SNR

- The probability of detection error P_e is usually found in terms of A (e.g. 3-ary constellation $\mathcal{C} = \{-2A, 0, 2A\}$).

We can find A in terms of average energy per bit E_b

↳ The average energy per symbol E_s of the constellation is given by

$$E_s = \sum_{c \in \mathcal{C}} P(x=c) |c|^2$$

For a M -PAM constellation, each symbol corresponds to $\log_2 M$ bits, so the average energy per bit E_b is

$$E_b = \frac{E_s}{\log_2 M} = \frac{\sum_{c \in \mathcal{C}} P(x=c) |c|^2}{\log_2 M}$$

→ substituting for A in the expression for P_e , we get P_e in terms of $\frac{E_b}{N_0} \propto \text{SNR}$.

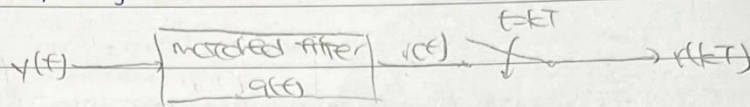
As before, it is useful to use the bound $Q(x) < \frac{1}{2} e^{-x^2/2}$ for large x .

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choosing the pulse $p(t)$ — Nyquist pulse criterion

criterion for choosing the pulse $p(t)$

- consider arbitrary $q(t)$ used in the matched filter structure.



We want to properly choose $p(t)$, $q(t)$ so we can perfectly recover the transmitted symbols from the sampled output of the matched filter when there is no noise.

- when there is no noise, the filter output $r(t)$ is given by

$$r(t) = x(t) * q(t) = \int_{-\infty}^{\infty} q(u) x(t-u) du = \sum_k x_k \int_{-\infty}^{\infty} q(u) p(t-kT-u) du = \sum_k x_k g(t-kT)$$

where $g(t) = q(t) * p(t) = \int_{-\infty}^{\infty} q(u) p(t-u) du$ is the effective pulse at the receiver.

($p(t)$: transmit filter; $q(t)$: receive filter; $g(t)$: overall filter)

- Sampling the filter output $r(t)$ at $t = mT$, for $m \in \mathbb{Z}$, we have

$$r(mT) = \sum_k x_k g((m-k)T)$$

If we want $r(mT) = x_m$ for all $m \in \mathbb{Z}$, then the overall filter $g(t)$ must satisfy

$$g(mT) = \begin{cases} 1 & \text{if } m=0 \\ 0 & \text{o/w} \end{cases}$$

o/w, we would have inter-symbol interference (ISI).

$$r(mT) = x_m + \text{ISI contributions from } \{x_k\}_{k \neq m}$$

Nyquist pulse criterion

- the above condition on $g(t)$ is equivalent to the Nyquist pulse criterion in the freq. domain.

$$\sum_{n=-\infty}^{\infty} G(f - \frac{n}{T}) = T$$

where $G(f) = \text{FT}[g(t)]$ is the FT of the effective pulse $g(t)$.

- consider multiplying the effective pulse $g(t)$ by an impulse train,

$$g(t) \left[\sum_{n=-\infty}^{\infty} \delta(t - nT) \right] = \sum_n g(nT) \delta(t - nT)$$

For no ISI, we req. $g(nT) = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{o/w} \end{cases}$, so $\sum_n g(nT) \delta(t - nT) = g(0) \delta(t - 0) = \delta(t)$.

$$g(t) \left[\sum_{n=-\infty}^{\infty} \delta(t - nT) \right] = \delta(t)$$

Taking FT on both sides, we get

$$G(f) + \text{FT} \left[\sum_{n=-\infty}^{\infty} \delta(t - nT) \right] = 1$$

Noting that $\text{FT} \left[\sum_{n=-\infty}^{\infty} \delta(t - nT) \right] = \text{FT} \left[\sum_n \frac{1}{T} e^{j2\pi n t/T} \right] = \sum_n \frac{1}{T} \delta(f - \frac{n}{T})$, we get

$$G(f) * \sum_{n=-\infty}^{\infty} \delta(f - \frac{n}{T}) = T \quad \rightarrow \quad \sum_n G(f - \frac{n}{T}) = T$$

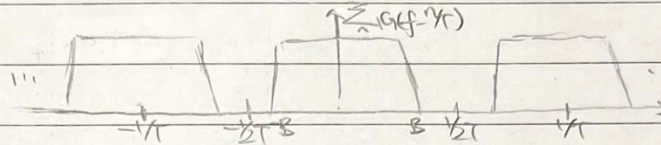
* Note that the proof goes both ways, i.e. $g(nT) = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{o/w} \end{cases} \Leftrightarrow \sum_n G(f - \frac{n}{T}) = T$

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Implications of the Nyquist pulse criterion

- The Nyquist pulse criterion implies that to not have ISI, $G(f)$ must have BW at least $1/2T$.

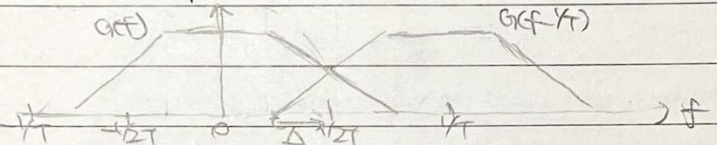
consider a $G(f)$ band-limited to $[-B, B]$, $B < \frac{1}{2T}$ as follows. $\rightarrow \sum_n G(f - n/T)$ has gaps so cannot uniformly equal T .



If $G(f)$ has BW greater than $1/4T$, then we are wasting BW by signaling too slowly

$\rightarrow$ we will consider the case where the pulse BW $B \in [1/4T, 1/2T)$.

- consider a real and even $G(f)$, i.e. $G(f) = G(f)$ (this req. $g(t)$ to be real and even).



Given the pulse BW $B \in [1/4T, 1/2T)$, we req. for all $\Delta \in [0, 1/4T)$

$$G(\frac{1}{2T} - \Delta) + G(\frac{1}{2T} + \Delta) = T \rightarrow G(\frac{1}{2T} - \Delta) + G(\frac{1}{2T} + \Delta) = T$$

This condition is the band-edge symmetry condition

The (root)-raised-cosine function

- In practice, the effective pulse $g(t)$ is often chosen to have a raised cosine transform

The raised cosine freq. function for a given rolloff factor $\beta \in [0, 1]$ is defined by

$$G_\beta(f) = \begin{cases} T \cos^2\left(\frac{\pi}{2\beta} \left(1 + \frac{(1-\beta)f}{1/4T}\right)\right) & 0 \leq |f| \leq \frac{1-\beta}{2T} \\ \frac{1}{4} & \frac{1-\beta}{2T} \leq |f| \leq \frac{1+\beta}{2T} \\ 0 & |f| > \frac{1+\beta}{2T} \end{cases}$$

- Recall that $g(t) = p(t) + z(t) \Leftrightarrow G(f) = P(f) + Q(f)$. We can choose the transmit and receive filters, $P(f)$ and $Q(f)$ as:

$$P(f) = Q(f) = \sqrt{G_\beta(f)}$$

where $\sqrt{G_\beta(f)}$ is the root raised cosine freq. function.

- The corresponding pulse $p(t)$ is given by the IFT of $P(f) = \sqrt{G_\beta(f)}$

$$p(t) = \frac{1}{\pi T} \left[\frac{\cos((1+\beta)\pi/4T) + \frac{(1-\beta)\pi}{4T} \sin((1-\beta)\pi/4T)}{1 - (4\beta^2/4T)^2} \right]$$

For $\beta=0$, $p(t)$ becomes the sinc pulse $p(t) = \frac{1}{T} \text{sinc}(\frac{t}{T})$, which decays as $\sim 1/t$.

For $0 < \beta \leq 1$, $p(t)$ decays as $\sim 1/t^2$ (faster than a sinc).

- The impulse response of the overall filter $g(t) = p(t) + z(t)$ is given by the IFT of $G(f) = G_\beta(f)$

$$g(t) = \frac{1}{T} \text{sinc}\left(\frac{t}{T}\right) \frac{\cos(\pi\beta/4T)}{1 - (4\beta^2/4T)^2}$$

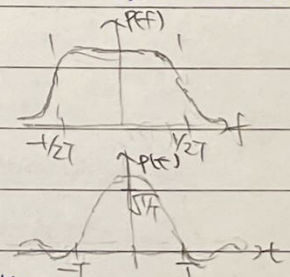
For $\beta=0$, $g(t)$ is a sinc, which decays as $\sim 1/t$.

For $0 < \beta \leq 1$, $g(t)$ decays as $\sim 1/t^2$ (much faster than a sinc).

* As β increases from 0 to 1, the pulse $p(t)$ decays faster but BW also increases from $1/4T$ to $1/2T$.

* $p(t), g(t)$ are real, even and have infinite support $\rightarrow p(t)$ is truncated & delayed to make it causal.

(the faster the decay in t (i.e. larger β), the less the deviation from Nyquist pulse criterion due to truncation)



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other sources of ISI.

- There are other factors that can introduce ISI (even when Nyquist pulse criterion is satisfied).

→ $g(t)$ does not exactly satisfy Nyquist pulse criterion (e.g. due to truncation of $p(t)$)

→ The sampling times are not exactly integer multiples of T . (e.g. we sample at $\{kT + \Delta\}_{k \in \mathbb{Z}}$)

Nyquist pulse criterion and orthogonality of pulse shifts.

- Let be $G(f)$ be any real non-negative function that satisfies the Nyquist pulse criterion $\sum_k G(f - \frac{k}{T}) = T$

Let the transmit and receive filters satisfy

$$|P(f)| = |Q(f)| = \sqrt{G(f)}$$

Then for $P(f)Q(f) = G(f)$, we req. $\angle Q(f) = -\angle P(f)$, i.e.

$$Q(f) = P^*(f)$$

As $p(t)$ is real, $P^*(f) = P(-f)$, therefore

$$Q(f) = P(-f)$$

→

$$q(t) = p(-t)$$

- w/ the choice $q(t) = p(-t)$, the overall impulse response $g(t)$ is

$$g(t) = p(t) * q(t) = p(t) * p(-t) = \int_{-\infty}^{\infty} p(u) p(u-t) du$$

As $g(t)$ satisfies the Nyquist pulse criterion, in the time domain, $g(nT) = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{o.w.} \end{cases}$, so

$$g(nT) = \int_{-\infty}^{\infty} p(u) p(u-nT) du = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{o.w.} \end{cases}$$

i.e. $\{p(t-nT)\}_{n \in \mathbb{Z}}$ is an orthonormal set of functions

→ w/ the matched filter $q(t) = p(-t)$, the orthonormal shift property of $p(t)$ is equivalent to

$G(f) = |P(f)|^2$ satisfying the Nyquist pulse criterion.

Power spectral density of PAM

The other model for the PAM signal.

- consider a slightly modified PAM signal (other model)

$$x(t) = \sum_k x_k p(t - kT - \theta)$$

where $\theta \sim \text{Unif}[0, T]$ is a random offset/delay.

- The other models how an observer measuring the PSD has no information abt. when the transmitter's clock was switched on, and it models $x(t)$ a WSS process.

- We will also assume that:

(i) x_k are drawn from a constellation w/ zero mean, e.g. $\{-3A, -A, A, 3A\}$.

(ii) $\{x_k\}_{k=-\infty}^{\infty}$ is WSS → $E[x_k] = 0$, $E[x_k^2] < \infty$, $r_{xx}[k_1, k_2] = r_{xx}[k_2 - k_1]$ for all k .

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Properties of the carrier model for the PAM signal

- For the carrier model of the PAM signal, $x_b(t) = \sum_k x_k p(t - kT - \theta)$

(i) The mean $E[x_b(t)]$ is zero, $E[x_b(t)] = 0$. x_k and θ independent

$$E[x_b(t)] = E\left[\sum_k x_k p(t - kT - \theta)\right] = \sum_k E[x_k] E[p(t - kT - \theta)] = 0$$

(ii) The autocorrelation $r_{x_b x_b}[\tau]$ is given by $r_{x_b x_b}[\tau] = \frac{1}{T} \sum_m r_{xx}[m] r_{pp}[\tau - mT]$ x_k and θ independent

$$r_{x_b x_b}[\tau] = E[x_b(t) x_b(t + \tau)] = E\left[\sum_k x_k p(t - kT - \theta) \sum_l x_l p(t + \tau - lT - \theta)\right] = \sum_k \sum_l E[x_k x_l] E[p(t - kT - \theta) p(t + \tau - lT - \theta)]$$

$$= \sum_k \sum_l r_{xx}[l - k] \left[\frac{1}{T} \int_0^T p(t - kT - \theta) p(t + \tau - lT - \theta) d\theta \right] \quad \leftarrow E[g(\theta)] = \int_0^T g(\theta) \frac{1}{T} d\theta = \int_0^T \frac{1}{T} g(\theta) d\theta$$

$$\begin{aligned} m = l - k &= \frac{1}{T} \sum_m r_{xx}[m] \sum_k \int_0^T p(t + mT - lT - \theta) p(t + \tau - lT - \theta) d\theta \\ &= \frac{1}{T} \sum_m r_{xx}[m] \int_0^T p(t + \tau - \theta) p(t + mT - \theta) d\theta \end{aligned} \quad \leftarrow \sum_k \int_0^T p(t - kT - \theta) d\theta = \int_0^T p(t) d\theta$$

$$v = t + \tau - \theta \quad = \frac{1}{T} \sum_m r_{xx}[m] \int_0^T p(v) p(v + mT - \tau) dv = \frac{1}{T} \sum_m r_{xx}[m] r_{pp}[\tau - mT]$$

where we define $r_{pp}[\tau] = \int_0^T p(v) p(v - \tau) dv$

(iii) The PSD $S_{x_b}(f)$ is given by $S_{x_b}(f) = \frac{|P(f)|^2}{T} \sum_m r_{xx}[m] e^{-j2\pi m f T}$

Note that $r_{pp}[\tau] = \int_0^T p(v) p(v - \tau) dv = r_{pp}[\tau] + r_{pp}[-\tau] \rightarrow FT[r_{pp}[\tau]] = P(f) P^*(f) = |P(f)|^2$

$$S_{x_b}(f) = \frac{1}{T} \sum_m r_{xx}[m] FT[r_{pp}[\tau - mT]] = \frac{1}{T} \sum_m r_{xx}[m] |P(f)|^2 e^{-j2\pi m f T} = \frac{|P(f)|^2}{T} \sum_m r_{xx}[m] e^{-j2\pi m f T}$$

✓ (i), (ii) → carrier model for PAM signal is WSS.

- In most applications, the symbols $\{x_k\}$ are independent as independent sets of bits are mapped to successive constellation symbols. In this case $r_{xx}[m]$ is given by

$$r_{xx}[m] = \begin{cases} E[x_k^2] = E_s & \text{if } m=0 \\ 0 & \text{o.w.} \end{cases}$$

where E_s is the average energy per symbol.

- The expressions for autocorrelation $r_{x_b x_b}[\tau]$ and PSD $S_{x_b}(f)$ simplify to

$$r_{x_b x_b}[\tau] = \frac{E_s}{T} r_{pp}[\tau] \quad S_{x_b}(f) = \frac{E_s}{T} |P(f)|^2$$

- The average power of the PAM waveform is therefore

$$\begin{aligned} E[x_b(t)^2] &= \int_{-\infty}^{\infty} S_{x_b}(f) df = \frac{E_s}{T} \int_{-\infty}^{\infty} |P(f)|^2 df \\ E[x_b(t)^2] &= r_{x_b x_b}[0] = \frac{E_s}{T} r_{pp}[0] = \frac{E_s}{T} \int_{-\infty}^{\infty} |p(t)|^2 dt \end{aligned} \quad \leftarrow \text{Parseval's theorem} \quad \int_{-\infty}^{\infty} |p(t)|^2 dt = \int_{-\infty}^{\infty} |P(f)|^2 df$$

If the pulse $p(t)$ has unit energy, the power of PAM signal is $P_{PAM} = \frac{E_s}{T}$

- If the matched filter is chosen as $q(t) = p(-t)$, then $G(f) = P(f) P^*(f) = |P(f)|^2$, and the PSD $S_{x_b}(f)$ is given by

$$S_{x_b}(f) = \frac{E_s}{T} G(f)$$

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Passband modulation - QAM and PSK

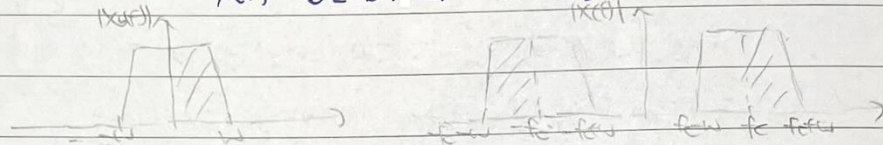
Upconverted PAM

- We can up-convert a PAM signal from baseband to passband by multiplying by a high freq. carrier

$$x(t) = x_b(t) \cos(2\pi f_c t) = \left[\sum_k x_k p(t - kT) \right] \cos(2\pi f_c t)$$

- The spectrum of the passband signal $x(t)$ is given by

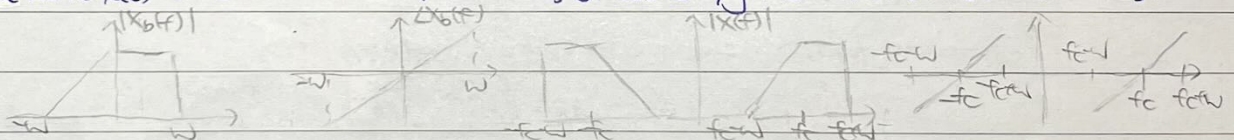
$$X(f) = \frac{1}{2} [X_b(f - f_c) + X_b(f + f_c)]$$



- In PAM, both the pulse $p(t)$ and the information symbols $\{x_k\}$ are real-valued, $\rightarrow$ the spectrum of the baseband signal $x_b(t)$ satisfies $X_b(f) = X_b^*(f) \rightarrow$ spectrum $X_b(f)$ for $f < 0$ is completely determined by the spectrum for $f > 0 \rightarrow$ inefficient use of BW.
- PAM signal has BW W , passband signal has BW $2W \rightarrow$ we are transmitting information at the same rate as PAM, but using twice the BW.

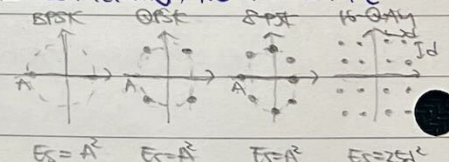
Quadrature amplitude modulation (QAM)

- We want $x(t)$ to be a real-valued passband signal but carrying information on both side bands.



- In QAM, we still use the same baseband waveform $x_b(t)$, but now the constellation from which the symbols x_k are drawn can be complex-valued

$$x_b(t) = \sum_k x_k p(t - kT)$$



We req. the transmitted passband waveform $x(t)$ to be real, so

$$x(t) = \text{Re}[\sqrt{2} X_b(t) e^{j2\pi f_c t}] = \text{Re}[X_b(t)] \sqrt{2} \cos(2\pi f_c t) - \text{Im}[X_b(t)] \sqrt{2} \sin(2\pi f_c t) \\ = \sum_k p(t - kT) [\text{Re}(x_k) \sqrt{2} \cos(2\pi f_c t) - \text{Im}(x_k) \sqrt{2} \sin(2\pi f_c t)]$$

- where $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ are the in-phase and quadrature components respectively
- + $\sqrt{2}$ factor makes the carriers have unit power.

Orthormal basis for QAM

- Define the following functions for $k \in \mathbb{Z}$:

$$f_k^c(t) = p(t - kT) \sqrt{2} \cos(2\pi f_c t)$$

$$f_k^s(t) = -p(t - kT) \sqrt{2} \sin(2\pi f_c t)$$

The transmitted QAM waveform can be written as

$$x(t) = \sum_k [X_k^c f_k^c(t) + X_k^s f_k^s(t)]$$

where $X_k^c = \text{Re}[x_k]$ and $X_k^s = \text{Im}[x_k]$

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- $\{f_k(t), \tilde{f}_k(t)\}_{k \in \mathbb{Z}}$ forms an orthonormal basis, provided that $p(t)$ is a baseband pulse w/ BW W and is orthogonal to $p(t-nT)$ for non-zero integers n

$$\langle f_k, f_l \rangle = \begin{cases} 1 & \text{if } k=l \\ 0 & \text{o.w.} \end{cases} \quad \langle \tilde{f}_k, \tilde{f}_l \rangle = \begin{cases} 1 & \text{if } k=l \\ 0 & \text{o.w.} \end{cases} \quad \langle f_k, \tilde{f}_l \rangle = 0$$

(proved by expanding double angle formula, and showing $\int_{-\infty}^{\infty} p(t-kT)p(t-lT)\cos(2\pi f_k t)dt = 0$ by applying the multiplication theorem and considering that the spectra $P(f), P(f \pm 2f_k)$ do not have overlapping supports).

- The QAM signal is $x(t) = \sum_k [X_k^I f_k(t) + X_k^Q \tilde{f}_k(t)] \rightarrow \mathbf{X}_k = \begin{bmatrix} X_k^I \\ X_k^Q \end{bmatrix}$ is the projection coeff of $x(t)$ w/ $\mathbf{f}_k(t) = \begin{bmatrix} f_k(t) \\ \tilde{f}_k(t) \end{bmatrix}$

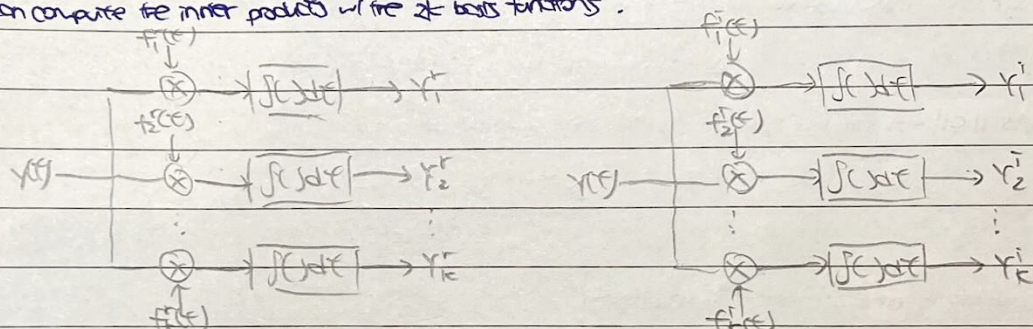
Let $\mathbf{N}_k = \begin{bmatrix} N_k^I \\ N_k^Q \end{bmatrix}$ and $\mathbf{Y}_k = \begin{bmatrix} Y_k^I \\ Y_k^Q \end{bmatrix}$ be the projection coeff of $n(t)$ and $y(t)$ w/ $\mathbf{f}_k(t) = \begin{bmatrix} f_k(t) \\ \tilde{f}_k(t) \end{bmatrix}$ respectively.

$$N_k^I = \langle n, f_k \rangle \quad N_k^Q = \langle n, \tilde{f}_k \rangle \quad Y_k^I = \langle y, f_k \rangle \quad Y_k^Q = \langle y, \tilde{f}_k \rangle$$

Note that we can write $\mathbf{Y}_k = \mathbf{X}_k + \mathbf{N}_k$ since

$$\mathbf{Y}_k = \begin{bmatrix} Y_k^I \\ Y_k^Q \end{bmatrix} = \begin{bmatrix} \langle y, f_k \rangle \\ \langle y, \tilde{f}_k \rangle \end{bmatrix} = \begin{bmatrix} \langle x+n, f_k \rangle \\ \langle x+n, \tilde{f}_k \rangle \end{bmatrix} = \begin{bmatrix} \langle x, f_k \rangle + \langle n, f_k \rangle \\ \langle x, \tilde{f}_k \rangle + \langle n, \tilde{f}_k \rangle \end{bmatrix} = \begin{bmatrix} X_k^I \\ X_k^Q \end{bmatrix} + \begin{bmatrix} N_k^I \\ N_k^Q \end{bmatrix} = \mathbf{X}_k + \mathbf{N}_k$$

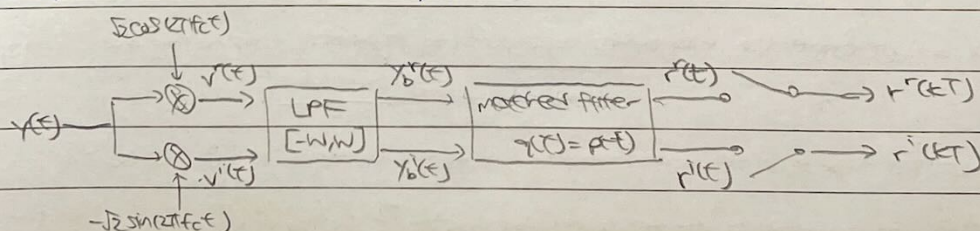
- To detect symbols $X_1, \dots, X_K$, the receiver can be implemented w/ a bank of K "correlators", which can compute the inner products w/ the $2K$ basis functions.



This implementation is not practical when we want to transmit a large amount of data (large K)

Carrier multiplication + LPF + matched filter implementation

- The QAM receiver can be constructed by product modulator, LPF and PAM receiver.



As the baseband pulse $p(t)$ is band limited to $[-W/2, W/2]$, the LPF removes high freq components, so at the LPF o/p,

$$y(t) = \sum_k X_k^I p(t-kT) + n(t)$$

$$y(t) = \sum_k X_k^Q p(t-kT) + n(t)$$

The problem has been reduced to detecting $\mathbf{X}_k = \begin{bmatrix} X_k^I \\ X_k^Q \end{bmatrix}$ from the PAM-like received waveform $\mathbf{y}(t) = \begin{bmatrix} y^I(t) \\ y^Q(t) \end{bmatrix}$

- The carrier multiplication + LPF followed by a matched filter is an efficient implementation of the optimal signal space receiver.

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Optimal detection for QAM.

- We want to detect the symbol $\underline{x}_k = \begin{bmatrix} x_k^I \\ x_k^Q \end{bmatrix}$ from $\underline{y}_k = \underline{x}_k + \underline{n}_k$ for $k \in \mathbb{Z}$, where $\underline{x}_k$ is one of $\underline{c} \in \mathcal{C}$ where $N_1, N_1, N_2, N_2 \dots$ i.i.d. $N(0, N/2)$ since $\underline{n}(t)$ is white Gaussian noise w/ PSD $N/2$

- The optimal detection rule is the MAP rule. When $\underline{y} = \underline{y}$,

$$\hat{\underline{x}}_{\text{MAP}} = \underset{\underline{c} \in \mathcal{C}}{\text{argmax}} P(\underline{x} = \underline{c} | \underline{y}) = \underset{\underline{c} \in \mathcal{C}}{\text{argmax}} P(\underline{x} = \underline{c}) P(\underline{y} | \underline{x} = \underline{c})$$

If all the symbols in the constellation are equally likely, i.e. $P(\underline{x} = \underline{c})$ is const. for all $\underline{c} \in \mathcal{C}$, then we have the ML rule

$$\hat{\underline{x}}_{\text{ML}} = \underset{\underline{c} \in \mathcal{C}}{\text{argmax}} P(\underline{y} | \underline{x} = \underline{c})$$

- For AWGN, $N_1, N_1 \sim N(0, N/2)$ and $\underline{y}^I = \underline{x}^I + \underline{n}^I$, $\underline{y}^Q = \underline{x}^Q + \underline{n}^Q$, so for $\underline{x} = \underline{c}$, $\underline{y}^I \sim N(\underline{c}^I, N/2)$, $\underline{y}^Q \sim N(\underline{c}^Q, N/2)$, i.e.

$$P(\underline{y} | \underline{x} = \underline{c}) = \frac{1}{(\pi N/2)^2} \exp\left(-\frac{(\underline{y}^I - \underline{c}^I)^2}{N/2}\right) \frac{1}{(\pi N/2)^2} \exp\left(-\frac{(\underline{y}^Q - \underline{c}^Q)^2}{N/2}\right) = \frac{1}{(\pi N)^2} \exp\left(-\frac{(\underline{y} - \underline{c})^2}{N}\right)$$

The ML rule is therefore

$$\hat{\underline{x}}_{\text{ML}} = \underset{\underline{c} \in \mathcal{C}}{\text{argmin}} (\underline{y}^I - \underline{c}^I)^2 + (\underline{y}^Q - \underline{c}^Q)^2 = \underset{\underline{c} \in \mathcal{C}}{\text{argmin}} \|\underline{y} - \underline{c}\|^2 = \underset{\underline{c} \in \mathcal{C}}{\text{argmin}} \|\underline{c}\|^2 - 2\underline{y}^T \underline{c}$$

- Note for PSK, we can define $\theta(\underline{c}, \underline{y})$ to be the angle between vectors $\underline{c}$ and $\underline{y}$. $\|\underline{y} - \underline{c}\|^2$ is given by

$$\|\underline{y} - \underline{c}\|^2 = \|\underline{c}\|^2 + \|\underline{y}\|^2 - 2\|\underline{c}\|\|\underline{y}\|\cos(\theta(\underline{c}, \underline{y}))$$

As $\|\underline{c}\| = \|\underline{y}\| = A$ for all symbols in PSK, the ML rule is

$$\hat{\underline{x}}_{\text{ML}} = \underset{\underline{c} \in \mathcal{C}}{\text{argmin}} \theta(\underline{c}, \underline{y})$$

Probability of detection error P_e and SNR

- The probability of detection error P_e is usually found in terms of A (e.g. BPSK $\mathcal{C} = \{A, -A\}$)

We can find A in terms of average energy per bit E_b

↳ The average energy per symbol E_s of the constellation is given by

$$E_s = \sum_{\underline{c} \in \mathcal{C}} P(\underline{x} = \underline{c}) \|\underline{c}\|^2$$

For a M-ary constellation, each symbol corresponds to $\log_2 M$ bits, so the average energy per bit E_b is

$$E_b = \frac{E_s}{\log_2 M} = \frac{\sum_{\underline{c} \in \mathcal{C}} P(\underline{x} = \underline{c}) \|\underline{c}\|^2}{\log_2 M}$$

→ Substituting for A in the expression for P_e , we get P_e in terms of $\frac{E_b}{N_0} \propto \text{SNR}$

* As before, it is useful to use the bound $Q(x) < \frac{1}{2} e^{-x^2/2}$ for large x .

Connection with PAM detection error probability

- For square QAM constellations, we effectively have PAM in two dimensions.

- Denoting P_c as probability for correct detection,

correct detection of real part

$$P_c^{\text{M-QAM}} = P_c^{\text{PAM}} \cdot P_c^{\text{PAM}} = (1 - P_e^{\text{PAM}})^2$$

correct detection of imaginary part

correct detection of QAM

$$P_e^{\text{M-QAM}} = 1 - P_c^{\text{M-QAM}} = 1 - (1 - P_e^{\text{PAM}})^2$$

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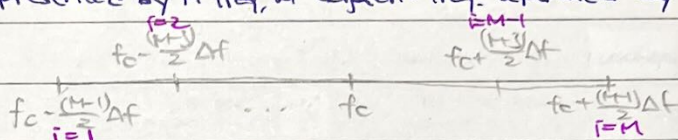
Frequency Shift Keying (FSK)

- In FSK, the information modulates the freq. of the carrier. In M-ary FSK, to transmit message $i \in \{1 \dots M\}$ in any symbol period $[lT, (l+1)T)$, the FSK waveform $x(t)$ is

$$x(t) = \sqrt{E_s} \sum_{i=1}^M \cos(2\pi (f_c + (2i - (M+1)) \frac{\Delta f}{2}) t)$$

where $\Delta f = \frac{1}{2T} = \frac{f_0}{K}$ for some large integer K .

- The M symbols are represented by M freq. w/ adjacent freq. separated by.



Orthogonal basis for FSK.

- Define the following functions for $i=1 \dots M$, $t \in [lT, (l+1)T)$ where $l \in \mathbb{Z}$.

$$f_i(t) = \sqrt{\frac{E_s}{T}} \cos(2\pi (f_c + (2i - (M+1)) \frac{\Delta f}{2}) t)$$

The transmitted FSK waveform can be written as

$$x(t) = \sqrt{E_s} f_i(t)$$

- $\{f_i(t) \dots f_M(t)\}$ forms an orthogonal basis. Assume that $t \in [0, T)$ (similar for the case $t \in [lT, (l+1)T)$)

$$\langle f_i, f_j \rangle = \int_0^T f_i(t) f_j(t) dt = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} \langle f_i, f_j \rangle &= \int_0^T f_i(t) f_j(t) dt = \int_0^T \sqrt{\frac{E_s}{T}} \cos(2\pi (f_c + (2i - (M+1)) \frac{\Delta f}{2}) t) \cos(2\pi (f_c + (2j - (M+1)) \frac{\Delta f}{2}) t) dt \\ &= \frac{1}{T} \int_0^T [\cos(2\pi (2f_c + ((i+j) - (M+1)) \Delta f) t) - \cos(2\pi (i-j) \Delta f t)] dt \end{aligned}$$

The first term integrates to $\frac{1}{2T} \sin(2\pi (2f_c + ((i+j) - (M+1)) \Delta f) T)$, which equals zero since

$$2f_c T = K, \quad 2\Delta f T = 1 \rightarrow 2\pi (2f_c + ((i+j) - (M+1)) \Delta f) T = \pi [2(2f_c T) + ((i+j) - (M+1)) (2\Delta f T)]$$

integer

The second term integrates to 1 if $i=j$, o.w. it integrates to $-\frac{1}{2T} \sin(2\pi (i-j) \Delta f T)$, which equals zero.

- The FSK signal is $x(t) = \sum_k x_k f_k(t) = \sqrt{E_s} f_i(t) \rightarrow \mathbf{x} = [x_1 \dots x_i \dots x_M] = [0 \dots \sqrt{E_s} \dots 0]$ is the projection coeff vector of $x(t)$ w/ the orthogonal basis $\{f_1(t) \dots f_M(t)\}$.

Let N_k and Y_k be the projection coeff of $n(t)$ and $y(t)$ w/ $f_k(t)$ respectively

$$N_k = \langle n, f_k \rangle$$

$$Y_k = \langle y, f_k \rangle$$

Note that we can write $Y_k = X_k + N_k$ since

$$Y_k = \langle y, f_k \rangle = \langle x + n, f_k \rangle = \langle x, f_k \rangle + \langle n, f_k \rangle = X_k + N_k$$

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Optimal detection for FSK

- As before, $N_1, N_2, \dots$ iid $\sim N(0, N_0/2)$ since $n(t)$ is white Gaussian noise w/ PSD $N_0/2$

If message $i \in \{1, \dots, M\}$ is transmitted,

$$x_k = \begin{cases} \sqrt{E_b} & \text{if } k=i \\ 0 & \text{o.w.} \end{cases} \rightarrow [Y_1, \dots, Y_i, \dots, Y_M] = [N_1, \dots, E_b + N_i, \dots, N_M]$$

- The optimal detection rule is the MAP rule. When $Y = y$

$$\hat{m}_{\text{MAP}} = \underset{i \in \{1, \dots, M\}}{\operatorname{argmax}} P(X = x(i) | Y) = \underset{i \in \{1, \dots, M\}}{\operatorname{argmax}} P(X = x(i)) P(Y | X = x(i))$$

If all the messages are equally likely, i.e. $P(X = x(i))$ is const for all $i \in \{1, \dots, M\}$, then we have the ML rule

$$\hat{m}_{\text{ML}} = \underset{i \in \{1, \dots, M\}}{\operatorname{argmax}} P(Y | X = x(i))$$

- For AWGN, $N \sim N(0, N_0/2)$ and $Y = X + N$, so for $X = x(i)$, $Y = [Y_1, \dots, Y_i, \dots, Y_M] = [N_1, \dots, E_b + N_i, \dots, N_M]$, i.e.

$$p(Y | X = x(i)) = P(Y_1 | X_1 = 0) \dots P(Y_i | X_i = \sqrt{E_b}) \dots P(Y_M | X_M = 0) = \left(\frac{1}{\sqrt{N_0/2}}\right)^M \exp\left(-\frac{(Y_1^2 + \dots + Y_M^2)}{N_0} - \frac{E_b}{N_0}\right) \exp\left(\frac{2\sqrt{E_b} Y_i}{N_0}\right)$$

The ML rule is therefore

$$\hat{m}_{\text{ML}} = \underset{i \in \{1, \dots, M\}}{\operatorname{argmax}} Y_i$$

Probability of detection error P_e and SNR

- Due to symmetry, the probability of error is the same regardless of which message $i \in \{1, \dots, M\}$

was transmitted $\rightarrow$ we can assume message 1 was transmitted w/o loss of generality.

- An error occurs if y_i is not the max. along $y = [Y_1, \dots, Y_M]$, so

$$P_e = P\left(\bigcup_{k=2}^M \{Y_i \leq Y_k\}\right) = P\left(\bigcup_{k=2}^M \{\sqrt{E_b} + n_i \leq n_k\}\right) \leq \sum_{k=2}^M P(\sqrt{E_b} + n_i \leq n_k) = (M-1)P(n_2 - n_1 \geq \sqrt{E_b})$$

since $(n_2 - n_1), (n_3 - n_1), \dots, (n_M - n_1)$ are each RVs $\sim N(0, \frac{N_0}{2} + \frac{N_0}{2} = N_0)$, we can thus express using $Q(x)$,

$$P_e = (M-1)P\left(\frac{n_2 - n_1}{\sqrt{N_0}} \geq \frac{\sqrt{E_b}}{\sqrt{N_0}}\right) = (M-1)Q\left(\frac{\sqrt{E_b}}{\sqrt{N_0}}\right) \leq \frac{1}{2}(M-1)e^{-\frac{E_b \log 2 M}{2 N_0}} = \frac{1}{2}e^{-\frac{\log 2 M}{2} \frac{E_b}{N_0}} e^{\log 2 M \ln 2} \leq \frac{1}{2}e^{-\frac{\log 2 M}{2} \frac{E_b}{N_0}}$$

$$\therefore P_e \leq \frac{1}{2} \exp\left[-\frac{\log 2 M}{2} \left(\frac{E_b}{N_0} - 2 \ln 2\right)\right]$$

Therefore if $\frac{E_b}{N_0} > 2 \ln 2$, the probability of detection error P_e decreases as M increases.

$\rightarrow$ Intuitively, this is possible because we do not decrease the distance between signals when increasing M .

Rate and bandwidth — QAM vs FSK

- Both M -ary QAM and M -ary FSK have the rate $\frac{\log_2 M}{T}$ bits/s

- For QAM, the BW is determined by the bandpass pulse $p(t)$. If $p(t)$ is bandlimited to $[L, W]$, then

the QAM signal $x(t)$ is bandlimited to $[f_c - W, f_c + W]$, so has BW $2W$

1 For $p(t)$ to satisfy the Nyquist pulse criterion, $W \in [\frac{1}{2T}, \frac{1}{T})$

- For M -FSK, given message $i \in \{1, \dots, M\}$ is transmitted, the signal $x(t)$ is a cosine w/ freq. $f_c + (i-1)\Delta f$

Therefore the total BW req. for M -FSK is $(M-1)\Delta f = \frac{M-1}{2T}$ (since $\Delta f = \frac{1}{2T}$)

- We define the BW efficiency η as $\eta = \frac{\text{rate}}{\text{BW}}$

assume BW $\neq 0$

$$\eta_{\text{QAM}} \approx \log_2 M \text{ bits/s/Hz}$$

$$\eta_{\text{FSK}} = \frac{2 \log_2 M}{M-1} \text{ bits/s/Hz}$$

* As M increases, $\eta_{\text{QAM}} \uparrow$ but $\eta_{\text{FSK}} \downarrow$ whereas $P_{e, \text{QAM}} \uparrow$ but $P_{e, \text{FSK}} \downarrow \rightarrow$ trade-off between η , P_e

channel equalisation

Frequency dispersive channels

- So far we considered channels whose frequency response $H(f)$ was flat throughout the transmission band

→ we can use the additive noise model $y(t) = x(t) + n(t)$.

- However, we may have a channel freq. response $H(f)$ that is not flat in the transmission band.

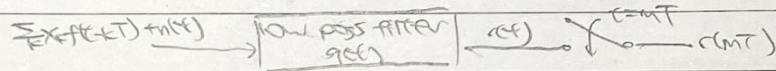
(freq. selective/dispersive channel). In general, the channel o/p $y(t)$ is given by

$$y(t) = \int h(u) x(t-u) du + n(t)$$

- Consider the PAM signal $x(t) = \sum_k x_k p(t-kT)$. The received signal $y(t)$ is

$$y(t) = \int h(u) x(t-u) du + n(t) = \sum_k x_k f(t-kT) + n(t), \text{ where } f(t) = p(t) * h(t)$$

At the receiver, we have



In the absence of noise, $r(t) = \sum_k x_k g(t-kT)$, where the overall filter $g(t)$ is

$$g(t) = f(t) * q(t) = p(t) * h(t) * q(t)$$

* There will be no ISI (i.e. $r(mT) = x_m$) if the overall filter $g(t)$ satisfies the Nyquist pulse criterion.

- The o/p of the demodulator $r(mT)$ is given by

$$r(mT) = \sum_k x_k g((m-k)T) = \dots + x_{m-2} g(2T) + x_{m-1} g(T) + x_m g(0) + x_{m+1} g(-T) + \dots \text{ for } m=0,1,\dots$$

Denoting $r_m = r(mT)$ and $g_m = g(mT)$, we write the above eqn. as

$$r_m = \sum_k g_k x_{m-k} \text{ for } m=0,1,\dots$$

$$\text{i.e. } \{r_m\} = \{g_m\} * \{x_m\}$$

- In practice, only a finite no. of entries in g are non-negligible. Assume the only non-zero entries

in g are $[g_0, \dots, g_L] = [g(0), \dots, g(LT)]$. The above equation becomes

$$r_m = \sum_{k=0}^L g_k x_{m-k} \text{ for } m=0,1,\dots$$

Taking Z-transform on both sides, we have

$$R(z) = G(z) X(z)$$

Zero-forcing equaliser

- If we pass the demodulator o/p r through a filter w/ transfer function $H_E(z) = \frac{1}{G(z)}$, the filter o/p is

$$Y(z) = R(z) H_E(z) = G(z) X(z) \cdot \frac{1}{G(z)} = X(z)$$

i.e. we recover the sequence of information symbols x perfectly → $H_E(z) = \frac{1}{G(z)}$ is the zero-forcing filter.

- In the presence of noise, the o/p of the ZF filter is

$$Y(z) = [G(z) X(z) + N(z)] H_E(z) = X(z) + \frac{N(z)}{G(z)}$$

Taking the inverse Z-transform, we have the sequence

$$y_m = x_m + \tilde{n}_m \quad m=0,1,\dots, \text{ where } \tilde{n}_m \text{ is the inverse Z-transform of } \frac{N(z)}{G(z)}.$$

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- Note that $G(z) = \sum_{l=0}^{\infty} g_l z^{-l}$, the transfer function of the ZF filter is

$$H(z) = \frac{1}{\sum_{l=0}^{\infty} g_l z^{-l}} = \frac{1}{g_0 + g_1 z^{-1} + g_2 z^{-2} + \dots}$$

finite no. of terms
infinite no. of terms

therefore the above ZF filter is an IIR filter.

- Consider the Z-transform of the ZF filter o/p y , $Y(z)$

$$Y(z) = H(z) R(z) = \frac{R(z)}{G(z)} \rightarrow R(z) = Y(z) G(z) \rightarrow r_n = \sum_{l=0}^{\infty} g_l y_{n-l}$$

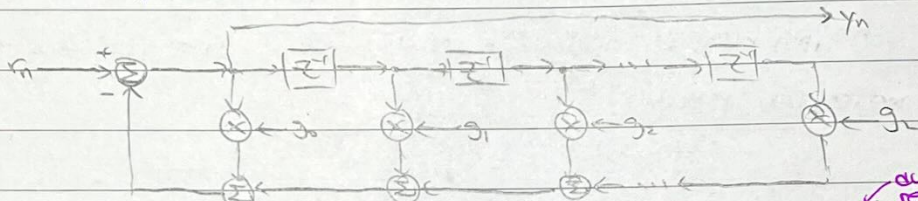
expanding the sum and rearranging,

$$r_n = g_0 y_n + g_1 y_{n-1} + \dots + g_L y_{n-L}$$

o/p of matched filter post ZF filter o/p

$$\rightarrow y_n = \frac{1}{g_0} [r_n - g_1 y_{n-1} - \dots - g_L y_{n-L}]$$

We can thus implement the IIR ZF filter as follows:



IIR filters are difficult to deal w in practice - stability is not guaranteed

Zero-forcing equaliser (FIR)

- Consider an equalising filter w/ $K+1$ taps, w/ impulse response $h = [h_0, \dots, h_K]$

recall the denominator o/p r is given by $r = g * x + n$, so the equalising filter o/p is

$$y = h * r = (h * g) * x + h * n = f * x + n$$

where $f = h * g$.

Note that $f_j = \sum_{i=0}^K h_i g_{j-i}$ for $0 \leq j \leq L+K$ ← since $h = [h_0 \dots h_K]$
 $g = [g_0 \dots g_L]$

- Therefore, the o/p of the FIR ZF filter is

$$y_m = x_m f_0 + \sum_{j=1}^{L+K} x_{m-j} f_j + \sum_{i=0}^K h_i n_{m-i}$$

symbolic noise L+K symbols of ISI noise convolved w FIR ZF filter

For an ideal ZF equaliser, we would like

$$f_0 = 1, \quad f_j = 0 \text{ for } j=1, 2, \dots, L+K.$$

where

$$\begin{aligned} f_0 &= h_0 g_0 \\ f_1 &= h_0 g_1 + h_1 g_0 \\ &\vdots \\ f_K &= h_0 g_K + h_1 g_{K-1} + \dots + h_K g_0 \\ &\vdots \\ f_{L+K} &= h_K g_L \end{aligned}$$

L+K+1 equations, but only K+1 filter coeff.

→ we cannot simultaneously satisfy $L+K+1$ eqns by specifying only $K+1$ filter coeff → perfect

zero-forcing w/ FIR filter is not possible.

- we typically choose $K \geq L$, then determine $b = [b_0, \dots, b_K]$ by solving.

$$f_0 = 1, \quad f_j = 0 \text{ for } j=1, 2, \dots, K$$

constraining the first K eqns constrain the effect of the strongest interfering coeffs.

- The o/p of the FIR ZF filter w/ $K+1$ taps is

$$y_m = x_m f_0 + \sum_{j=1}^{L+K} x_{m-j} f_j + \sum_{i=0}^K h_i n_{m-i} = x_m + \sum_{j=1}^{L+K} x_{m-j} f_j + \sum_{i=0}^K h_i n_{m-i}$$

Assuming the noise variables $\{n_i\}$ are iid $N(0, \sigma^2)$, the o/p noise variance is given by

$$E\left[\left(\sum_{i=0}^K h_i n_{m-i}\right)^2\right] = \sigma^2 \left(\sum_{i=0}^K h_i^2\right)$$

← noise enhanced by factor of $\sum h_i^2$

→ As the no filter taps $K+1$ increases, we reduce the residual interference but increase noise variance at o/p.

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MMSE equalisation

- The ZF approach does not take the noise enhancement effect into account when designing the filter. The MMSE equaliser explicitly tries to min. the expected square error b/w x_n and its estimate $\hat{x}_n$.
- For a $k+1$ tap MMSE equaliser, for each n , the estimate $\hat{x}_n$ is generated as

$$\hat{x}_n = c_0 r_n + c_1 r_{n+1} + \dots + c_k r_{n+k} = \mathbf{r}^T \mathbf{c}$$

where r_n is the demodulator o/p at $t=nT$, $\mathbf{c} = \begin{bmatrix} c_0 \\ \vdots \\ c_k \end{bmatrix}$, $\mathbf{r} = \begin{bmatrix} r_n \\ \vdots \\ r_{n+k} \end{bmatrix}$

- * Note the equaliser can be implemented as a filter w/ impulse response $[h_0=c_0, h_1=c_1, \dots, h_k=c_k]$, where the $(k+1)$ th symbol of this filter o/p equals $\hat{x}_n$.

- We want to determine the vector $\mathbf{c} = [c_0 \dots c_k]$ that minimises $E[(x_n - \hat{x}_n)^2]$,
→ not explicitly cancel out the ISI, but find best estimate of $\hat{x}_n$ to x_n .

$$f(\mathbf{c}) = E[(x_n - \mathbf{r}^T \mathbf{c})^2] \rightarrow \frac{\partial f(\mathbf{c})}{\partial c_j} = 2E[r_{nj}(x_n - \mathbf{r}^T \mathbf{c})] = 0 \text{ for } j=0,1,\dots,k.$$

These set of $k+1$ eqns can be collectively written as

$$E[\mathbf{r}(x_n - \mathbf{r}^T \mathbf{c})] = \mathbf{0} \rightarrow E[\mathbf{r}x_n] = E[\mathbf{r}\mathbf{r}^T] \mathbf{c}$$

→ The MMSE equaliser is $\mathbf{c} = \mathbf{R}^{-1} \mathbf{p}$, where $\mathbf{R} = E[\mathbf{r}\mathbf{r}^T]$, $\mathbf{p} = E[\mathbf{r}x_n]$

- * Note that both $\mathbf{R}$ and $\mathbf{p}$ are deterministic.

Remarks on ZF and MMSE equalisers.

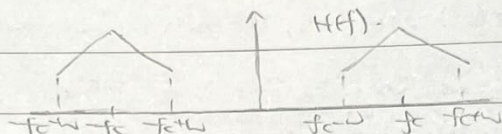
- Both the ZF and MMSE equalisers are linear equalisers — the estimate of each symbol is a linear combination of the observed demodulator o/p $\{r_n\}$.
- Among the linear equalisers, ZF equaliser achieves the min. residual ISI, MMSE equaliser achieves the min. expected square error per symbol.
- The same equalisation techniques can be applied to deal w/ ISI in either baseband channels w/ PAM or passband channels w/ QAM. (For QAM, $x_n \in \mathbb{C}$).
- We assumed a fixed set of coeff. for the overall filtering. In practice, the channel may be time-varying → equalization filter coefficients have to be updated accordingly (adaptive equalisation).
- Equalisation is useful to eliminate ISI regardless of how it arises.

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Orthogonal Frequency Division Multiplexing (OFDM)

transmitted waveform and receiver structure.

- Consider the freq. dispersive channel w/ the following frequency response $H(f)$



suppose $x_b(t)$ is formed using N symbols as follows:

$$x_b(t) = \sum_{m=0}^{N-1} x[m] p(t-mT) \quad \text{In OFDM, } x[m] \text{ is not chosen from QAM constellation.}$$

If $x_b(t)$ is the transmitted waveform for the baseband equivalent channel, for the passband channel, we transmit

$$x(t) = \operatorname{Re}[x_b(t)e^{j2\pi f_c t}]$$

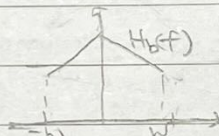
Ignoring additive noise $n(t)$, the received waveform is

$$y(t) = x(t) * h(t) = \sum_{m=0}^{N-1} [\operatorname{Re}[x[m]] \cos(2\pi f_c t) - \operatorname{Im}[x[m]] \sin(2\pi f_c t)] p(t-mT)$$

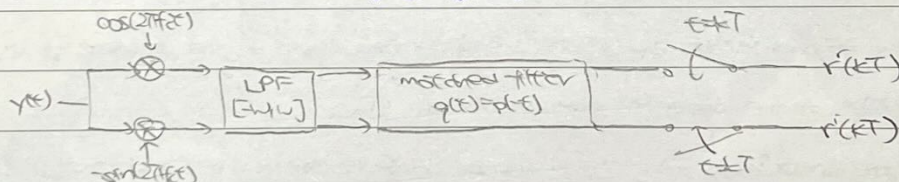
where $f(t) = p(t) * h_b(t)$,

- $h_b(t)$ is the baseband equivalent impulse response, $h(t) = \operatorname{Re}[h_b(t)e^{j2\pi f_c t}]$

the baseband equivalent frequency response of the channel $H_b(f)$ is



- the standard receiver structure is as follows:



the noiseless baseband equivalent signal after matched filtering is

$$r(t) = \sum_{m=0}^{N-1} x[m] g(t-mT) \quad \text{where } g(t) = \text{channel impulse response} = p(t) * h_b(t) + n(t) \quad \text{TX filter, RX filter}$$

the sampled match filter o/p is therefore

$$r[k] = r(kT) = \sum_{m=0}^{N-1} x[m] g[k-m] = \sum_{l=0}^{L-1} g[l] x[k-l] \quad \text{assume } g[l] = g(L) \text{ not zero only for } 0 \leq l \leq L-1 \quad \text{for } k=0, 1, \dots, N-1$$

Linear and circular convolution

- Linear and circular convolutions are defined as follows. For $k=0, 1, \dots, N-1$,

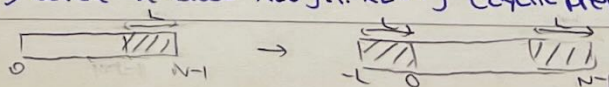
$$r[k] = \sum_{l=0}^{L-1} g[l] x[k-l] \quad \tilde{r}[k] = \sum_{l=0}^{L-1} g[l] x[(k-l) \bmod N]$$

- Notice that $L < N$, for $k=L, L+1, \dots, N-1$, $r[k] = \tilde{r}[k]$. However, for $k=0, 1, \dots, L-1$,

$$r[k] = g[0]x[k] + g[1]x[k-1] + \dots + g[k]x[0] + g[k+1]x[N-1] + g[k+2]x[N-2] + \dots + g[L]x[N-L+k]$$

$$\tilde{r}[k] = g[0]x[k] + g[1]x[k-1] + \dots + g[k]x[0] + g[k+1]x[N-1] + g[k+2]x[N-2] + \dots + g[L]x[N-L+k]$$

- To make $r[k]$ mimic $\tilde{r}[k]$, we define $x[-L] = x[N-L] \dots x[-1] = x[N-1]$ and transmit the symbols $x[-L] \dots x[-1]$ before the block $x[0] \dots x[N-1]$ cyclic prefix.



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OFDM transmitter and receiver

The OFDM transmitter works as follows

1) Encode the information in DFT domain by choosing $x[0] \dots x[N-1]$ from a QAM constellation.

2) Take the inverse DFT to produce the time domain sequence $x[0] \dots x[N-1]$

3) Insert the cyclic prefix $\{x[L] \dots x[L-1]\} = \{x[N-L] \dots x[N-1]\}$ $\leftarrow \text{IDFT: } x[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{j2\pi kn/N}$

4) Using the time-domain sequence, form $x_b(t) = \sum_{n=0}^{N-1} x[n] p(t - nT)$

5) Transmit the passband signal $x(t) = \text{Re}\{x_b(t) e^{j2\pi f_c t}\}$

Each OFDM block has total duration $(L+N)T$ seconds and transmits N QAM symbols.

$\leftarrow \text{rate} = \frac{N}{(L+N)T}$ QAM symbols/second.

The OFDM receiver works as follows

1) Pass the received signal $y(t) = x(t) * h(t) + n(t)$ through the standard receiver structure (multiplication + LPF + matched filter)

$\leftarrow$ equivalent to circular convolution as we added a cyclic prefix

2) Sample the matched filter op at $t = kT$ $r[k] = \sum_{l=0}^{L-1} g[l] x[k-l] + n[k]$ for $k = L \dots 0 \dots N-1$

Discard the ops $r[L] \dots r[0]$. $\leftarrow \text{DFT: } x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{j2\pi kn/N}$

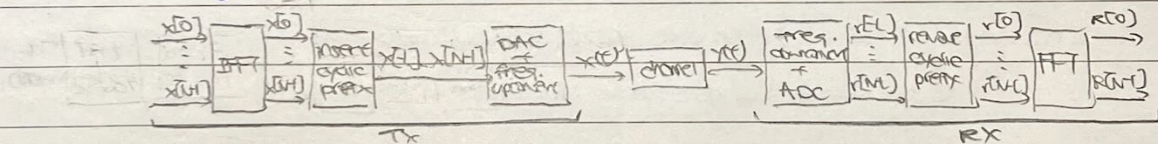
3) Compute the N -pt DFT of $r[0] \dots r[N-1]$ to get $R[n] = G[n] x[n] + N[n]$ for $n = 0 \dots N-1$

4) Use MAP rule to recover the QAM symbols $x[n]$

$\leftarrow$ effectively a AWGN channel w/ no ISI

We can find $G[0] \dots G[N-1]$ by taking the N -pt DFT of $g[0] \dots g[N-1]$.

The block diagram of Tx + Rx implementation is as follows:



Overhead due to cyclic prefix

Each OFDM block has total duration $(L+N)T$ seconds. The period of duration LT to send the cyclic prefix is the guard interval/period. (occupies $\frac{L}{L+N}$ of the OFDM block).

The guard interval decreases the effective rate by a factor of $\frac{L}{L+N}$, but is essential to ensure no ISI.

Average energy

Denote the average energy of the QAM constellation by E_s (freq. domain constellation).

The average energy of the time-domain symbols is

$$E[|x_b[k]|^2] = E\left[\left(\frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{j2\pi kn/N}\right) \left(\frac{1}{N} \sum_{l=0}^{N-1} x[l] e^{j2\pi kl/N}\right)^*\right] = \frac{1}{N} \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} E[x[n] x[l]^*] e^{j2\pi k(n-l)/N}$$

For independent symbols, $E[x[n] x[l]^*] = \begin{cases} E_s & \text{if } l=n \\ 0 & \text{if } l \neq n \end{cases} \therefore E[|x_b[k]|^2] = E_s$

Although the average energy is unchanged, the peak energy of the time-domain constellation is greater than the freq. domain constellation $\rightarrow$ power amplifiers may be outside their safe operating region.

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Subcarrier interpretation of OFDM

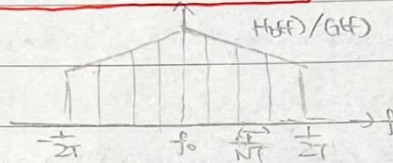
- Recall that the sampled match filter output is given by

$$R[n] = G[n]X[n] + N[n], \text{ for } n = 0, \dots, N-1$$

The DFT coeffs $\{G[n]\}$ correspond to samples of the spectrum $G(f)$ of N equally spaced freq. b/wn $-\frac{1}{2T}$ and $\frac{1}{2T}$.

- $G(f)$ is the FT of the overall baseband filter $G(f) = P(f)H_b(f)P^*(f) = (P(f))^2 H_b(f) \propto H_b(f)$.

For $0 \leq n \leq N-1$, we have $G[n] = G(f_n) = |P(f_n)|^2 H_b(f_n)$ where $f_n = \begin{cases} \frac{1}{T} \frac{n}{N} & 0 \leq n < N/2 \\ \frac{1}{T} \frac{(N-n)}{N} & N/2 \leq n \leq N-1 \end{cases}$



- If the pulse $P(f)$ is roughly const (i.e. $P(f) \approx c$) for $-\frac{1}{2T} \leq f \leq \frac{1}{2T}$, then

$$R[n] = c^2 H_b(f_n) X[n] + N[n] \text{ for } n = 0, \dots, N-1$$

We can interpret OFDM as having N subcarriers of freq $f_0 \dots f_{N-1}$, w/ spacing of $\frac{1}{NT}$ b/wn carriers.

- The QAM symbol $x[n]$ is carried by the sub-carrier f_n . When passed through a freq. dispersive channel, each sub-carrier f_n just sees a scalar channel gain $H_b(f_n)$.

(OFDM effectively converts the wideband dispersive channel into N non-interfering narrowband channels).

- The SNR for the channel corresponding to the n th carrier depends on the channel gain $G[n] \propto H_b(f_n)$.

- In coded OFDM, we can use an outer channel code for error correction (esp. for low SNR subcarriers).

We can also vary the symbol rate across subcarriers $\rightarrow$ higher rate for subcarriers w/ higher channel gain.

Bandwidth and rate of OFDM.

- We have N subcarriers, w/ spacing b/wn adjacent ones $\frac{1}{NT}$. In the passband, the sub-carrier freq. are $\{f_c + f_0, f_c + f_1, \dots, f_c + f_{N-1}\}$.

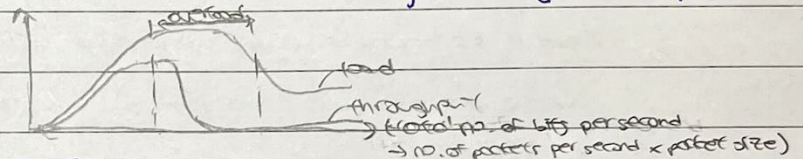
- The channel BW is given by $BW = \frac{1}{T}$

- The rate of transmission is $\text{rate} = \frac{N}{(NT)T}$ QAM symbols/second

Transmission Control Protocol (TCP) and congestion control

TCP and congestion

- The transport layer of a communication network ensures reliable end-to-end data transmission. Sources can use FB from destination to retransmit lost packets / adjust data transmission rates.
- TCP is used in the transport layer for (i) connection establishment, (ii) connection maintenance (reliability, congestion / flow control, sequencing) and (iii) connection termination.
- When too many sources send too much data too fast for the network to handle, we have congestion. This could lead to packet loss, retransmission $\rightarrow$ reduced throughput / congestion collapse.



$\rightarrow$ Congestion may continue after the overload

- The TCP approach to congestion control is as follows: *we want high network utilization, high throughput, avoid congestion collapse and fair BW sharing.*

$\rightarrow$ No explicit FB from network, but is a closed-loop end-to-end control system

$\rightarrow$ Congestion inferred from end-system observed packet loss and delay (no direct measurements)

TCP-Reno

- In TCP-Reno, the source sends packets to the receiver. The receiver responds w/ an acknowledgment ack every time a packet is received.

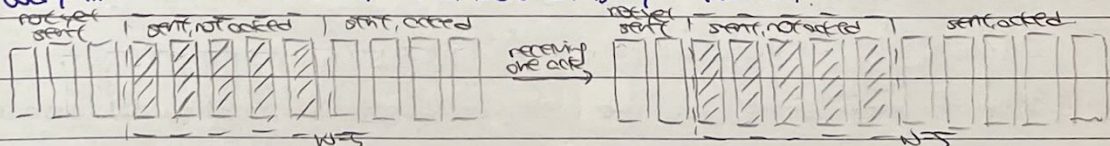
- Denote the round-trip time (RTT) b/w packet being sent and ack received w/ T .

Making repeated measurements of time from packet transmission until ack receipt, we calculate the empirical mean μ and standard deviation σ^2 . Set $T = \mu$ and $\text{timeout} = \mu + k\sigma$

- We assume a packet is lost if time elapsed $t > \text{timeout}$

- The source maintains a window containing a varying no. of packets W . *$\leftarrow$ regulates transmission speed.*

- Every time an ack is received, the window shifts by 1 position.



- An ack contains the index of the next packet the receiver expects.

$\rightarrow$ Given packet 3.2 received, upon receiving packet 3, the next ack asks for packet 4.

If then packet 5 is received, the next ack asks for packet 4 again.

- If two acks ask for the same packet, we have a duplicate acknowledgment / "dupack".

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Initialisation, slow start and congestion avoidance phase.

① Initialisation

- During initialisation, set the window size $W=1$, estimate the RTT T and select a slow start threshold value $ssthresh$ for W .

- During normal operation, when each ack is received, we send one or more packets whilst ensuring that the no. of unacknowledged packets is no greater than W .

(At time c , the rate $R(c) = \frac{W(c)}{T}$ packets/second.

② Slow start phase

- During the slow start phase, we increase W by 1 w/ each ack, when all goes well:

→ send $W=1$ packet → receive ack → set $W=2$

→ send $W=2$ packets → receive 2 acks → set $W=4$ etc.

- When $W > ssthresh$, we enter the congestion avoidance phase

③ Congestion avoidance phase

- During the congestion avoidance phase, we increase W by $\frac{1}{W}$ w/ each ack (i.e. increase W by 1 every T seconds). When all goes well:

→ send W packets → receive W acks → set $W=W+1$

→ send $W+1$ packets → receive $W+1$ acks → set $W=W+2$ etc.

- continue until we get delay (3x dupacks) or loss (timeout).

Fast retransmit and fast recovery

① Packet delay → fast retransmit

- The transmitter detects delay if it receives 3x dupacks for the same data → assumes the packet was irreparably delayed → resends missing packet before timeout

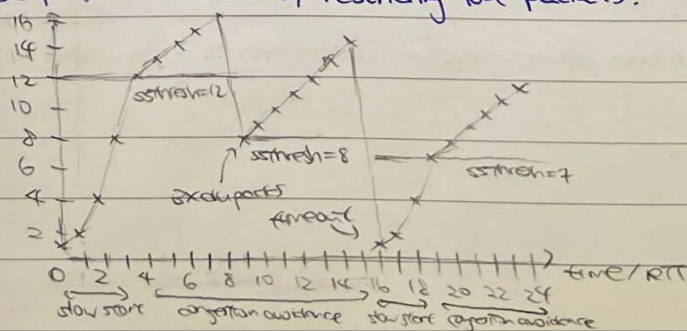
- Set $ssthresh$ to $W/2$ and the window size to $W/2$

② Packet loss → fast recovery

- The transmitter detects loss if it receives no acks for any of the packets sent from the window by timeout → assumes the packets are permanently lost.

- Set $ssthresh$ to $W/2$ and the window size to 1.

- Return to the slow start phase and start by resending lost packets.



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TCP-Reno operation (summary).

- TCP-Reno has a variable window size to regulate transmission speed.
- During slow-start, we have multiplicative rate increase until window size W reaches $ssthresh$.
During congestion avoidance, we have additive rate increase until delay/loss detected.
- Fast transmit when delay detected; Fast recovery when loss detected.
- Loss (timeout) is treated as a more serious issue than delay (3x dupacks). In both cases, we have a multiplicative rate decrease.
- In the long run, TCP-Reno spends almost all of the time in the congestion avoidance phase.
(additive increase multiplicative decrease (AIMD) adaptation)
- If the long run $ssthresh$ is k , then the average no. of packets is $\frac{k+k+1+\dots+2k}{k+1} = \frac{3k}{2}$

Continuous-time model for the equilibrium rate

- Assume we are in the congestion avoidance phase, and consider a continuous time approx model.

Let $q(t)$ be the loss rate at time t , and $R(t) = \frac{W(t)}{T}$ be the rate.

- The rate at which acks are received is $R(t-T)(1-q(t))$. Each ack increases $W(t)$ by $\frac{1}{W(t)}$
 packets sent at $t=T$
 Packet loss rate is $R(t-T)q(t)$. Each loss event decreases $W(t)$ by a factor of $\beta = 1/2$

$$\therefore \frac{dW(t)}{dt} = \frac{R(t-T)(1-q(t))}{W(t)} - \beta R(t-T)q(t)W(t)$$

$$\frac{dR(t)}{dt} = \frac{R(t-T)(1-q(t))}{T \cdot R(t)} - \beta R(t-T)q(t)R(t)$$

Solving $\frac{dR(t)}{dt} = 0$ gives $R(t) = \frac{1}{T\sqrt{\beta}} \sqrt{\frac{1-q(t)}{q(t)}}$ [TCP square root law]

- When the loss probability, $q(t)$ is small, then $R(t) \approx \frac{1}{T\sqrt{\beta q(t)}}$

* If the long run $ssthresh$ is k , we send a total of $\frac{3k}{2} \times k$ packets before a delay. $\rightarrow q(t) = \frac{1}{3k/2}$

Internet protocol (IP) and routing algorithms

The network routing problem

- A network is a directed graph $G = (N, L)$, where N is the set of nodes and L is the set of directed links $(i, j) \in N$. Each link $(i, j) \in L$ has an associated cost $c_{ij} > 0$.
- For a given source node u and each destination node i , the goal is to find the route $(u, j_1, \dots, j_m, i)$
Write min. cost $C_{uj_1} + C_{j_1j_2} + \dots + C_{j_mj_i} + C_{j_iu}$.
- We are given either
 - $\hookrightarrow$ The network topology and all link costs c_{ij} are known to each node [Link state algorithms]
 - $\hookrightarrow$ Each node knows its neighbors and the cost of the links to them [Distance vector algorithm]

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Dijkstra's algorithm

- Dijkstra's algorithm is an iterative, greedy algorithm that determines the paths $u \rightarrow i \forall i \in N$
- the min. total cost w_{ui}^* . (Finds min cost paths for a given node to all other nodes in the network)
- Dijkstra's algorithm is a link-state routing algorithm - the complete network topology and all costs must be known by everyone. It also requires $c_{ij} > 0$.
- Dijkstra's algorithm is as follows:

(i) Initialization

↳ fix the source node u .

↳ let the visited set $K \subset N$ be the nodes i for which mincost path $u \rightarrow i$ is already known.

↳ for each node i , let w_{ui} be the mincost $u \rightarrow i$ at current location

Initially set $w_{uu} = 0, w_{ui} = \infty$ for $i \neq u$.

↳ for each node i , let p_{ui} be the previous hop $\rightarrow i$ in current iteration min-cost path.

Initially set $p_{ui} = -1$ (unknown) for $i \neq u$.

(ii) Iterative step

↳ set the current node i^* to the source (i.e. $i^* = u$) and $K = \{u\}$

↳ while $K \neq N$:

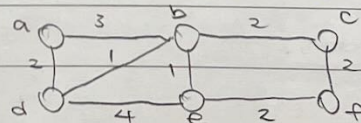
for each i satisfying $i \notin K$ and $(i^*, i) \in L$,

if $w_{ui} \leq w_{ui^*} + c_{i^*i}$ do nothing, else set $w_{ui} = w_{ui^*} + c_{i^*i}$ and $p_{ui} = i^*$

Choose a node $i^* \notin K$ achieving the min current cost, i.e. $i^* = \arg \min_{i \notin K} w_{ui}$

Add i^* to K

- eg: Consider the following graph. Find the min-cost path from the source a to other nodes.



Iteration	K	(w_{ab}, p_{ab})	(w_{ac}, p_{ac})	(w_{ad}, p_{ad})	(w_{ae}, p_{ae})	(w_{af}, p_{af})
1	{a}	(3, a)	(∞ , -1)	(2, a)	(∞ , -1)	(∞ , -1)
2	{a, d}	(3, a)	(∞ , -1)	(2, a)	(6, d)	(∞ , -1)
3	{a, d, b}	(3, a)	(5, b)		(4, b)	(∞ , -1)
4	{a, d, b, e}		(5, b)		(4, b)	(6, e)
5	{a, d, b, e, c}		(5, b)			(6, e)
6	{a, d, b, e, c, f}					(6, e)

↳ The actual min-cost path can be found by reading the table backwards.

- The algorithm always terminates after $|N| - 1$ iterations.

- In each iteration $1 \leq k \leq |N| - 1$, we increase the size of the visited set, $|K| = k + 1$

- The time complexity is $O(N^2)$ for naive algorithm, $O(N \log N + (L))$ for Fibonacci heap min-priority queue.

- Dijkstra's algorithm converges quickly

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Bellman-Ford algorithm

- Bellman-Ford algorithm is an iterative dynamic programming algorithm that determines the paths $u \rightarrow i \forall u, i \in N, i \neq u$ w/ min. total cost w_{ui} (finds all min cost paths b/w any two nodes)
- Bellman-Ford algorithm is a distributed, distance vector routing algorithm - only local information is used by each node.
- Bellman-Ford algorithm is as follows:

(i) Initialisation

- Each node $u \in N$ knows the cost c_{ui} for all its neighbouring nodes i , i.e. all $i \in N$ st. $(u,i) \in L$.
- Each node $u \in N$ maintains a distance vector (w_{ui}, n_{ui}) for each $i \in N$

w_{ui} is the min-cost along paths $u \rightarrow i$ of current iteration. Initially set $w_{ui} = 0, w_{ij} = \infty$ for $i \neq j$

n_{ui} is the next hop $\rightarrow i$ in current iteration min-cost path. Initially set $n_{ui} = -1$ (unknown) for $i \neq u$.

(ii) Iterative step

- FOR iterations $k=1, 2, \dots$

select a node $u \in N$ and for each $i \neq u$,

send all the current values (w_{ui}) to all the neighbours of u

broadcast cost to neighbouring nodes

repeat for all other nodes $u \in N$

select a node $u \in N$ and for each $i \neq u$,

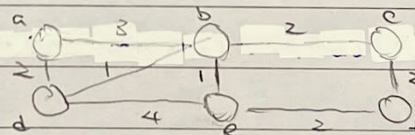
if adding a first hop $u \rightarrow j$ to the current $u \rightarrow i$ path helps, i.e. $\min_{j: (u,j) \in L} \{c_{uj} + w_{ji}\} < w_{ui}$,

set $w_{ui} = \min_{j: (u,j) \in L} \{c_{uj} + w_{ji}\}$ and $n_{ui} = \text{argmin}_{j: (u,j) \in L} \{c_{uj} + w_{ji}\}$

else do nothing.

repeat for all other nodes $u \in N$

- e.g.: consider the following graph. Find the min-cost paths from all nodes in the network to node a



Iteration	(w_{ab}, n_{ab})	(w_{ac}, n_{ac})	(w_{ad}, n_{ad})	(w_{ae}, n_{ae})	(w_{af}, n_{af})
1	(3, a)	(∞ , -1)	(2, a)	(∞ , -1)	(∞ , -1)
2	(3, a)	(5, b)	(2, a)	(4, b)	(∞ , -1)
3	(3, a)	(5, b)	(2, a)	(4, b)	(6, e)
4	(3, a)	(5, b)	(2, a)	(4, b)	(6, e)

+ The actual min-cost path can be found by reading the table backwards

- All optimal paths of length k will be found by the k th iteration

All optimal paths will be found by the $(N-1)$ th iteration

- The time complexity is $O(N|L|)$ (somewhat slower than Dijkstra's algorithm)

- Bellman-Ford can be implemented asynchronously - every node does their update in their own time and then sends the updated values to their neighbours when things change.

- The algorithm is sensitive to errors, link failures, changes in network topology $\rightarrow$ convergence can be slow after change.